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UPPER SIXTH — FURTHER MATHEMATICS

CHAPTER 15

APPLICATION OF SCALAR AND VECTOR PRODUCTS

SUGGESTED TIME: 6 PERIODS

IN THIS CHAPTER

This chapter applies the scalar (dot) and vector (cross) products to mechanics: resolving vectors along directions, computing work done by forces, moments of forces, couples, and the equilibrium of coplanar and three-dimensional force systems. It provides the essential toolkit for the GCE Advanced Level mechanics questions on statics.

- By the end of this chapter I can find the component (resolved part) of a vector in any given direction using the scalar product.
- By the end of this chapter I can compute the work done by a constant force along a straight displacement, including forces at an angle.
- By the end of this chapter I can calculate the moment of a force about a point in two and three dimensions using the vector product.
- By the end of this chapter I can apply the principle of moments to solve equilibrium problems involving levers, beams and ladders.
- By the end of this chapter I can identify and analyse a couple, compute its moment, and distinguish a couple from a single force.
- By the end of this chapter I can reduce a system of coplanar forces to a single resultant force or to a force and a couple.

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THIS SIGNED COURSE CONTAINS

COURSE complete, illustrated, step by step

METHODS and worked examples

EXERCISES exam style + solutions

SUMMARY SHEET the essentials to remember

CONTENTS

1. Section 1: Component of a Vector in a Given Direction (Lesson 102)
2. Section 2: Work Done by a Constant Force (Lesson 103)
3. Section 3: Moment of a Force and Equilibrium (Lesson 104)
4. Section 4: Single Forces and Couples (Lesson 105)
5. Section 5: Coplanar Forces and Simple 3D Systems (Lessons 106 & 107)
6. Integration activity
7. Methods and worked examples
8. Exercises
9. Solutions to the exercises
10. Summary sheet

CHAPTER OBJECTIVES

- By the end of this chapter I can find the component (resolved part) of a vector in any given direction using the scalar product.
- By the end of this chapter I can compute the work done by a constant force along a straight displacement, including forces at an angle.
- By the end of this chapter I can calculate the moment of a force about a point in two and three dimensions using the vector product.
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- By the end of this chapter I can identify and analyse a couple, compute its moment, and distinguish a couple from a single force.
- By the end of this chapter I can reduce a system of coplanar forces to a single resultant force or to a force and a couple.
- By the end of this chapter I can solve equilibrium problems for bodies under coplanar forces using resolution and moments.
- By the end of this chapter I can analyse simple three-dimensional force systems using vector methods (resultants and vector moments).

PREREQUISITES

- Scalar (dot) product and vector (cross) product of vectors, with their geometric interpretations (Lower Sixth).
- Vector algebra in 2D and 3D: components, magnitude, unit vectors, position vectors.
- Resolution of forces and basic notions of equilibrium from Lower Sixth mechanics.
- Trigonometry: sine, cosine, angles between lines, and solving simple trigonometric equations.
- Solving simultaneous linear equations in two or three unknowns.

1 SECTION 1: COMPONENT OF A VECTOR IN A GIVEN DIRECTION (LESSON 102)

On the Bamenda–Bafoussam road, a heavily loaded lorry stalls on a steep slope: the driver must know how much of the lorry's weight pulls it backwards along the road, and how hard the handbrake must work to hold it. At the Douala port, a crane operator tilts a container and wonders at what angle it will topple. In a mechanic's workshop in Yaoundé, two spanners are used together to loosen a rusted nut that a single spanner cannot move. All these situations — components of forces, work, turning effects, couples and equilibrium — are governed by the **scalar product** and the **vector product** you met in Lower Sixth. This chapter turns those algebraic tools into the language of mechanics.

Our first problem is simple to state but fundamental: given a vector $\vec{a}$ (say a force) and a direction (say a slope), how much of $\vec{a}$ acts along that direction? The answer is the **component** of a vector in a given direction, and it is the key that unlocks work, moments and equilibrium in the rest of the chapter.

1.1 Review of the Scalar Product

DEFINITION **Scalar product**

For two vectors $\vec{a}$ and $\vec{b}$ with angle θ ($0 \le \theta \le \pi$) between them, the **scalar product** (or dot product) is

$$\vec{a} \cdot \vec{b} = |\vec{a}|\,|\vec{b}| \cos\theta.$$

If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$, then

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3, \qquad |\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}.$$

KEY POINTS **Essential facts about the dot product**

- $\vec{a} \cdot \vec{b}$ is a scalar, not a vector.
- $\vec{a} \cdot \vec{b} = 0 \iff \vec{a} \perp \vec{b}$ (for non-zero vectors).
- $\vec{a} \cdot \vec{a} = |\vec{a}|^2$.
- $\vec{i} \cdot \vec{i} = \vec{j} \cdot \vec{j} = \vec{k} \cdot \vec{k} = 1$ and $\vec{i} \cdot \vec{j} = \vec{j} \cdot \vec{k} = \vec{k} \cdot \vec{i} = 0$.
- The angle between two vectors is obtained from $\cos\theta = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$.

1.2 The Scalar Component (Resolved Part)

Imagine the sun directly overhead. The shadow that a stick $\vec{a}$ casts on the ground along a line in the direction of $\vec{b}$ has length $|\vec{a}|\cos\theta$. This "shadow length" is exactly what the driver on the Bamenda road needs: the amount of $\vec{a}$ that lies along the direction $\vec{b}$.

DEFINITION **Scalar component of $\vec{a}$ in the direction of $\vec{b}$**

The **scalar component** (or **resolved part**) of $\vec{a}$ in the direction of $\vec{b}$ is

$$\text{comp}_{\vec{b}}(\vec{a}) = \vec{a}\cdot\hat{\vec{b}} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = |\vec{a}|\cos\theta,$$

where $\hat{\vec{b}} = \dfrac{\vec{b}}{|\vec{b}|}$ is the **unit vector** in the direction of $\vec{b}$.

The derivation is immediate: since $\hat{\vec{b}}$ has length 1,

$$\vec{a}\cdot\hat{\vec{b}} = |\vec{a}|\,|\hat{\vec{b}}|\cos\theta = |\vec{a}|\cos\theta.$$

Sign of the component. Because $\cos\theta$ is positive for $\theta < 90^\circ$, zero for $\theta = 90^\circ$ and negative for $\theta > 90^\circ$, the scalar component tells us not only how much of $\vec{a}$ lies along $\vec{b}$, but also in which sense: a negative component means $\vec{a}$ points partly opposite to $\vec{b}$.

1.3 The Vector Projection

The scalar component gives a length with a sign. Often we want the actual vector shadow of $\vec{a}$ along the line of $\vec{b}$: this is the **vector projection**.

DEFINITION **Vector projection of $\vec{a}$ on $\vec{b}$**

The **vector projection** of $\vec{a}$ on $\vec{b}$ is the vector of magnitude $\text{comp}_{\vec{b}}(\vec{a})$ pointing along $\hat{\vec{b}}$:

$$\text{proj}_{\vec{b}}(\vec{a}) = \left(\vec{a}\cdot\hat{\vec{b}}\right)\hat{\vec{b}} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\,\vec{b}.$$

Component perpendicular to a direction. Since $\vec{a}$ is the sum of its parts along and perpendicular to $\vec{b}$, the part of $\vec{a}$ **perpendicular** to $\vec{b}$ is

$$\vec{a} - \text{proj}_{\vec{b}}(\vec{a}).$$

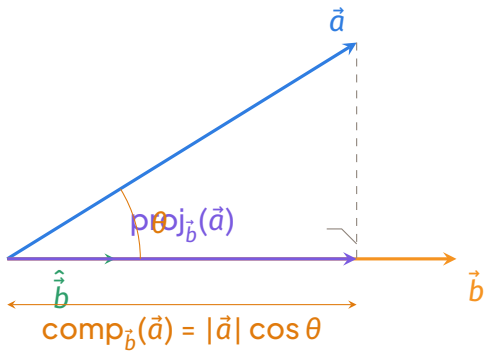
This vector is automatically perpendicular to $\vec{b}$: indeed $\left(\vec{a} - \text{proj}_{\vec{b}}(\vec{a})\right)\cdot\vec{b} = \vec{a}\cdot\vec{b} - \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}|\vec{b}|^2 = 0$.

WARNING **Scalar component versus vector projection**

The component $\text{comp}_{\vec{b}}(\vec{a}) = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$ is a scalar (it can be negative). The projection $\text{proj}_{\vec{b}}(\vec{a}) = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\vec{b}$ is a vector. Do not confuse them: examiners award separate marks for each; writing one when the question asks for the other loses them all.

WARNING **The most common exam error**

The formula $\text{proj}_{\vec{b}}(\vec{a}) = (\vec{a}\cdot\hat{\vec{b}})\hat{\vec{b}}$ contains $|\vec{b}|$ twice: once from the dot product with $\hat{\vec{b}}$ and once from multiplying by $\hat{\vec{b}}$. Forgetting to divide by $|\vec{b}|$ (or dividing only once) when $\vec{b}$ is not a unit vector is the single most frequent error at the GCE. Always check: is $\vec{b}$ already a unit vector? If not, divide by $|\vec{b}|$ for the component, and by $|\vec{b}|^2$ for the projection.



The vector projection of $\vec{a}$ on $\vec{b}$ (purple) and its scalar component, the signed length of the shadow of $\vec{a}$ along $\vec{b}$.

1.4 Components on the Coordinate Axes: Direction Cosines

A beautiful special case: take $\vec{b}$ to be one of the unit vectors $\vec{i},\vec{j},\vec{k}$. If $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$, then

$$\vec{a}\cdot\vec{i} = a_1, \qquad \vec{a}\cdot\vec{j} = a_2, \qquad \vec{a}\cdot\vec{k} = a_3.$$

So the Cartesian components of a vector are precisely its scalar components along the axes. If α,β,γ are the angles that $\vec{a}$ makes with the x-, y- and z-axes, then

$$\cos\alpha = \frac{a_1}{|\vec{a}|}, \qquad \cos\beta = \frac{a_2}{|\vec{a}|}, \qquad \cos\gamma = \frac{a_3}{|\vec{a}|}.$$

These are the **direction cosines** of $\vec{a}$, and they satisfy $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$ (since the unit vector $\hat{\vec{a}} = (\cos\alpha)\vec{i} + (\cos\beta)\vec{j} + (\cos\gamma)\vec{k}$ has length 1).

1.5 Resolving a Force Along a Given Line

METHOD **Resolving a force along a given line**

To find the resolved part of a force $\vec{F}$ along a line ℓ:

1. Find any vector $\vec{d}$ along the line ℓ (e.g. from two points on the line, or from the direction of an inclined plane).
2. Compute the unit vector $\hat{\vec{d}} = \dfrac{\vec{d}}{|\vec{d}|}$.
3. The resolved part of $\vec{F}$ along ℓ is the scalar $\vec{F} \cdot \hat{\vec{d}}$.
4. The force vector along ℓ is the projection $(\vec{F} \cdot \hat{\vec{d}})\,\hat{\vec{d}}$; the remainder $\vec{F} - (\vec{F} \cdot \hat{\vec{d}})\hat{\vec{d}}$ acts perpendicular to ℓ.
5. Interpret the sign: positive means $\vec{F}$ pulls along $\hat{\vec{d}}$; negative means it pulls against $\hat{\vec{d}}$.

WORKED EXAMPLE 1 **Component and projection with position vectors**

Statement. Given $\vec{a} = 2\vec{i} + 3\vec{j} + 6\vec{k}$ and $\vec{b} = 2\vec{i} - \vec{j} + 2\vec{k}$, find (a) the scalar component of $\vec{a}$ in the direction of $\vec{b}$; (b) the vector projection of $\vec{a}$ on $\vec{b}$; (c) the component of $\vec{a}$ perpendicular to $\vec{b}$; (d) the angle between $\vec{a}$ and $\vec{b}$.

SOLUTION

Solution. First compute

$$\vec{a} \cdot \vec{b} = (2)(2) + (3)(-1) + (6)(2) = 4 - 3 + 12 = 13,$$

$$|\vec{b}| = \sqrt{4 + 1 + 4} = 3, \qquad |\vec{a}| = \sqrt{4 + 9 + 36} = 7.$$

(a) Scalar component:

$$\operatorname{comp}_{\vec{b}}(\vec{a}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{13}{3}.$$

(b) Vector projection:

$$\operatorname{proj}_{\vec{b}}(\vec{a}) = \frac{13}{9}(2\vec{i} - \vec{j} + 2\vec{k}) = \frac{26}{9}\vec{i} - \frac{13}{9}\vec{j} + \frac{26}{9}\vec{k}.$$

(c) Perpendicular component:

$$\vec{a} - \operatorname{proj}_{\vec{b}}(\vec{a}) = \left(2 - \tfrac{26}{9}\right)\vec{i} + \left(3 + \tfrac{13}{9}\right)\vec{j} + \left(6 - \tfrac{26}{9}\right)\vec{k} = -\frac{8}{9}\vec{i} + \frac{40}{9}\vec{j} + \frac{28}{9}\vec{k}.$$

Check: $\left(-\tfrac{8}{9}\right)(2) + \left(\tfrac{40}{9}\right)(-1) + \left(\tfrac{28}{9}\right)(2) = \tfrac{-16-40+56}{9} = 0$, so it is indeed perpendicular to $\vec{b}$. ✓

(d) Angle:

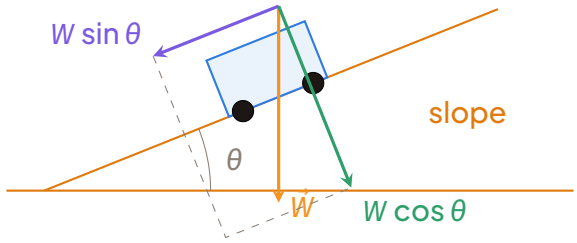
$$\cos\theta = \frac{13}{7 \times 3} = \frac{13}{21} \implies \theta \approx 51.8^{\circ}.$$

1.6 Application to Forces on an Inclined Plane

Return to our lorry on the Bamenda–Bafoussam road. Its weight $\vec{W}$ acts vertically downwards, but what matters for the handbrake is the resolved part of $\vec{W}$ along the slope. If the road makes an angle θ with the horizontal, a unit vector down the slope is $\hat{d} = (\cos\theta)\vec{i} - (\sin\theta)\vec{j}$ (taking $\vec{i}$ horizontal, $\vec{j}$ vertical), and with $\vec{W} = -W\vec{j}$:

$$\vec{W}\cdot\hat{d} = W\sin\theta.$$

So a force $W\sin\theta$ pulls the lorry down the road, while $W\cos\theta$ presses it into the road surface. This is why steep slopes demand so much more from brakes and engines.



The weight $\vec{W}$ of a lorry on a slope resolved into a component $W\sin\theta$ along the plane (which the brakes must balance) and $W\cos\theta$ perpendicular to the plane.

WORKED EXAMPLE 2 **Resolved part of a force along a line of action**

Statement. A force $\vec{F} = (3\vec{i} + 4\vec{j})$ N acts on a crate at a point O. A rope lies along the line from O to the point $A(4, 1)$ (coordinates in metres). Find (a) the resolved part of $\vec{F}$ along OA; (b) the force vector of $\vec{F}$ along OA; (c) the magnitude of the force perpendicular to the rope.

SOLUTION

Solution. (a) $\vec{OA} = 4\vec{i} + \vec{j}$, so $|\vec{OA}| = \sqrt{17}$ and

$$\hat{d} = \frac{1}{\sqrt{17}}(4\vec{i} + \vec{j}), \qquad \vec{F}\cdot\hat{d} = \frac{(3)(4) + (4)(1)}{\sqrt{17}} = \frac{16}{\sqrt{17}} \approx 3.88 \text{ N}.$$

(b) Force vector along OA:

$$\frac{16}{17}(4\vec{i} + \vec{j}) = \left(\frac{64}{17}\vec{i} + \frac{16}{17}\vec{j}\right) \text{ N}.$$

(c) Perpendicular force: $\vec{F} - \text{proj} = \left(3 - \frac{64}{17}\right)\vec{i} + \left(4 - \frac{16}{17}\right)\vec{j} = -\frac{13}{17}\vec{i} + \frac{52}{17}\vec{j}$, of magnitude

$$\frac{1}{17}\sqrt{13^2 + 52^2} = \frac{13\sqrt{17}}{17} = \frac{13}{\sqrt{17}} \approx 3.15 \text{ N}.$$

Check: $\left(\frac{16}{\sqrt{17}}\right)^2 + \left(\frac{13}{\sqrt{17}}\right)^2 = \frac{256+169}{17} = 25 = |\vec{F}|^2$. ✓

DID YOU KNOW? **Why this matters beyond the exam room**

Every time an engineer sizes a cable on a bridge, a physiotherapist estimates the force along a bone, or a pilot computes the headwind component on landing at Yaoundé–Nsimalen airport, the calculation is exactly $\vec{a}\cdot\hat{\vec{b}}$. The scalar component is arguably the most-used formula in all of applied vector mathematics.

EXERCISE 1 **Practice**

1. Given $\vec{a} = \vec{i} - 2\vec{j} + 2\vec{k}$ and $\vec{b} = 3\vec{i} + 4\vec{k}$, find the scalar component and the vector projection of $\vec{a}$ on $\vec{b}$.
2. A force $\vec{F} = (5\vec{i} - 2\vec{j} + \vec{k})$ N acts along a line in the direction of $\vec{d} = \vec{i} + 2\vec{j} - 2\vec{k}$. Find the resolved part of $\vec{F}$ along $\vec{d}$ and state whether it acts with or against $\vec{d}$.
3. A lorry of weight W stands on a road inclined at 12° to the horizontal. Find, in terms of W, the force the handbrake must provide to hold it at rest.
4. Find the direction cosines of $\vec{a} = 2\vec{i} - 3\vec{j} + 6\vec{k}$ and verify that their squares sum to 1.

SOLUTION **Exercise 1**

1. $\vec{a}\cdot\vec{b} = 3 + 0 + 8 = 11$, $|\vec{b}| = 5$. Component: $\frac{11}{5}$. Projection: $\frac{11}{25}(3\vec{i} + 4\vec{k})$.
2. $\vec{F}\cdot\vec{d} = 5 - 4 - 2 = -1$, $|\vec{d}| = 3$, so the resolved part is $-\frac{1}{3}$ N: it acts against $\vec{d}$.
3. $W\sin 12^\circ \approx 0.208\,W$.
4. $|\vec{a}| = 7$; direction cosines $\frac{2}{7}, -\frac{3}{7}, \frac{6}{7}$; squares: $\frac{4+9+36}{49} = 1$. ✓

KEY POINTS **Key points of Lesson 102**

- Scalar component: $\mathrm{comp}_{\vec{b}}(\vec{a}) = \vec{a}\cdot\hat{\vec{b}} = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$ — a signed scalar.
- Vector projection: $\mathrm{proj}_{\vec{b}}(\vec{a}) = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\vec{b}$ — a vector along $\vec{b}$.
- Perpendicular part: $\vec{a} - \mathrm{proj}_{\vec{b}}(\vec{a})$.
- Cartesian components are scalar components on $\vec{i}, \vec{j}, \vec{k}$; direction cosines are compo-

nents of the unit vector $\hat{\vec{a}}$.

- To resolve a force along a line: unit vector first, then dot product.

2 SECTION 2: WORK DONE BY A CONSTANT FORCE (LESSON 103)

In Section 1 we learnt how to split a vector into its **component** in a given direction and its component perpendicular to that direction. We saw that the component of a vector $\vec{F}$ along a direction $\vec{d}$ is $|\vec{F}|\cos\theta$, where θ is the angle between them. We now put this idea to work in one of the most important applications of the scalar product in mechanics: measuring the **work** done by a force. The key observation is simple: only the component of the force along the displacement actually does work. That component is exactly what the scalar product $\vec{F}\cdot\vec{d}$ captures.

2.1 Intuition: what does "work" mean in mechanics?

Imagine a porter at the Douala market pulling a loaded cart with a rope. The rope makes an angle θ with the horizontal ground, and the cart moves a distance $|\vec{d}|$ in a straight line. Everyday experience tells us two things:

- the harder the porter pulls (bigger $|\vec{F}|$), the more work is done;
- the farther the cart moves (bigger $|\vec{d}|$), the more work is done;
- pulling horizontally is more effective than pulling steeply upwards: when θ is large, much of the effort is wasted lifting the rope rather than moving the cart.

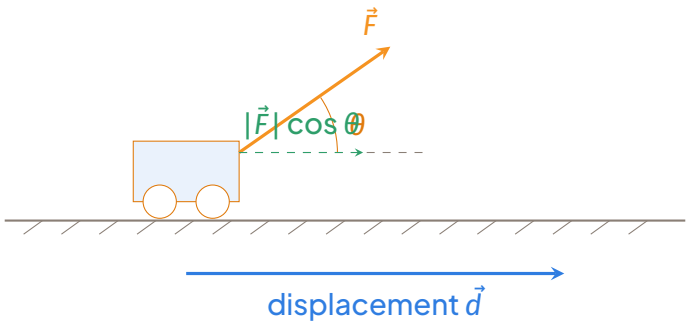
The quantity that combines these three observations is precisely $|\vec{F}||\vec{d}|\cos\theta$ — the scalar product of force and displacement.

DEFINITION **Work done by a constant force**

Let a **constant force** $\vec{F}$ act on a body which undergoes a straight-line displacement $\vec{d}$, and let θ be the angle between $\vec{F}$ and $\vec{d}$. The **work done** by the force is

$$W = \vec{F}\cdot\vec{d} = |\vec{F}|\,|\vec{d}|\cos\theta.$$

Work is a scalar quantity. Its SI unit is the **joule** (J): 1 J = 1 N m.



A porter pulls a cart with a force $\vec{F}$ at angle θ to the horizontal. Only the component $|\vec{F}|\cos\theta$ along the displacement does work.

Since $\vec{F}\cdot\vec{d} = (|\vec{F}|\cos\theta)|\vec{d}|$, the work is the product of the component of the force in the direction of motion (Section 1!) and the distance moved. Equivalently, $W = |\vec{F}|(|\vec{d}|\cos\theta)$: the force times the component of the displacement along the force. Both readings are useful.

2.2 Sign of the work: positive, negative and zero work

Because $W = |\vec{F}||\vec{d}|\cos\theta$, the sign of W is the sign of $\cos\theta$:

Angle θ	Sign of W	Physical meaning
$0^\circ \le \theta < 90^\circ$	$W > 0$	The force helps the motion (driving force).
$\theta = 90^\circ$	$W = 0$	The force neither helps nor hinders (e.g. normal reaction on a horizontal floor).
$90^\circ < \theta \le 180^\circ$	$W < 0$	The force opposes the motion (e.g. friction, braking force).

When a force does negative work, we often quote the work done **against** that force, which is the corresponding positive number. For example, if friction does work −150 J on a sliding crate, we say that 150 J of work is done against friction.

WARNING **By a force versus against a force**

Work done by friction is always negative (friction opposes motion); work done against friction is quoted as a positive number. Read the GCE question carefully to see which one is required, and give the sign accordingly.

2.3 Computing work from components

When the force and the displacement are given in component form, the dot product formula $W = \vec{F}\cdot\vec{d}$ is by far the fastest method — no angle is needed.

METHOD **Work from components**

1. Write the force as $\vec{F} = F_1\vec{i} + F_2\vec{j}$ (or $F_1\vec{i} + F_2\vec{j} + F_3\vec{k}$ in 3D).
2. Form the displacement vector $\vec{d} = \vec{AB} = \vec{b} - \vec{a}$, where $\vec{a}, \vec{b}$ are the position vectors of the start and end points A and B.
3. Compute $W = \vec{F} \cdot \vec{d} = F_1d_1 + F_2d_2$ $(+F_3d_3)$.
4. Attach the unit (joules) and interpret the sign.

WARNING **Displacement, not distance**

Work is defined using the displacement vector $\vec{d} = \vec{AB}$. If the path is not straight, do not use the total distance travelled. For a constant force, only the straight-line displacement from start to finish matters.

WORKED EXAMPLE 3 **Work done by a force in component form**

A force $\vec{F} = (3\vec{i} + 5\vec{j})$ N moves a particle from the point $A(2, 1)$ to the point $B(6, 4)$, where coordinates are in metres. Find the work done by $\vec{F}$.

SOLUTION

Solution. The displacement is

$$\vec{d} = \vec{AB} = (6 - 2)\vec{i} + (4 - 1)\vec{j} = 4\vec{i} + 3\vec{j} \text{ (m)}.$$

Hence

$$W = \vec{F} \cdot \vec{d} = (3)(4) + (5)(3) = 12 + 15 = 27 \text{ J}.$$

The work is positive: overall the force assists the motion.

WORKED EXAMPLE 4 **Porter pulling a cart on horizontal ground**

A porter pulls a cart along horizontal ground with a force of 80 N by means of a rope inclined at 30° to the horizontal. Find the work done by the pull when the cart moves 25 m in a straight line.

SOLUTION

Solution. Here $|\vec{F}| = 80$ N, $|\vec{d}| = 25$ m and $\theta = 30^\circ$:

$$W = |\vec{F}||\vec{d}|\cos\theta = 80 \times 25 \times \cos 30^\circ = 2000 \times \frac{\sqrt{3}}{2} = 1000\sqrt{3} \approx 1732 \text{ J}.$$

Note that the vertical component $80\sin 30^\circ$ does no work, since the cart does not move vertically.

2.4 Work done by gravity

Let a body of mass m move through a vertical displacement h (taking $h > 0$ upwards). The weight $\vec{W}_g = m\vec{g}$ acts vertically downwards with magnitude mg.

- If the body moves **up** through height h: the angle between weight and displacement is 180°, so $W_{\text{gravity}} = -mgh$. We say that work mgh is done against gravity.
- If the body moves **down** through height h: the angle is 0°, so $W_{\text{gravity}} = +mgh$.

A remarkable fact, consistent with the previous warning: since gravity is constant and vertical, the work done by gravity depends only on the vertical change in height, not on the path. A climber who ascends 10 m vertically along a winding mountain path has mgh of work done against gravity, exactly as if he had been lifted straight up.

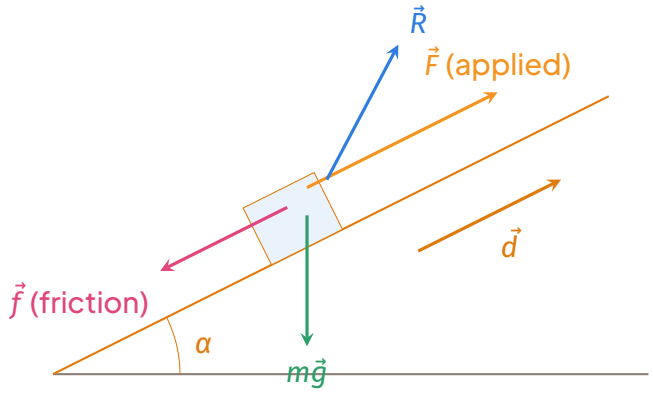
2.5 Total work of several forces

PROPERTY **Work of a resultant**

Work is a scalar, and the scalar product is distributive over vector addition. Consequently, if forces $\vec{F}_1, \vec{F}_2, \ldots, \vec{F}_n$ all act on a body during the same displacement $\vec{d}$, then the total work is the sum of the individual works, and equals the work of the resultant:

$$W_{\text{total}} = \sum_{i=1}^{n} \vec{F}_i \cdot \vec{d} = \left(\sum_{i=1}^{n} \vec{F}_i\right) \cdot \vec{d} = \vec{R} \cdot \vec{d}.$$

This is extremely useful on an inclined plane, where applied force, weight, friction and normal reaction all act at once. The normal reaction, being perpendicular to the plane, does no work.



A block pulled up a rough inclined plane: applied force $\vec{F}$, weight $m\vec{g}$, friction $\vec{f}$ (down the slope), normal reaction $\vec{R}$ (no work), displacement $\vec{d}$ up the slope.

WORKED EXAMPLE 5 **Total work on a rough inclined plane**

A crate of mass 20 kg is pulled 8 m up a plane inclined at 25° to the horizontal by a force of 150 N parallel to the plane. The frictional resistance is 40 N. Take $g = 9.8\ \text{m s}^{-2}$. Find (a) the work done by the applied force, (b) the work done by friction, (c) the work done by gravity, (d) the total work done on the crate.

SOLUTION

Solution. Take the displacement $\vec{d}$ of magnitude 8 m up the slope.
(a) The applied force is parallel to $\vec{d}$:

$$W_F = 150 \times 8 \times \cos 0^\circ = 1200\ \text{J}.$$

(b) Friction acts down the slope, opposite to $\vec{d}$:

$$W_f = 40 \times 8 \times \cos 180^\circ = -320\ \text{J}.$$

(c) The crate rises vertically through $h = 8 \sin 25^\circ$ m. Gravity acts downwards, so

$$W_g = -mgh = -20 \times 9.8 \times 8 \sin 25^\circ \approx -662.7\ \text{J}.$$

(Equivalently, the angle between the weight and $\vec{d}$ is $90^\circ + 25^\circ = 115^\circ$, and $W_g = 196 \times 8 \times \cos 115^\circ$, giving the same result.)
(d) The normal reaction does no work, so

$$W_{\text{total}} = 1200 - 320 - 662.7 = 217.3\ \text{J}.$$

The positive total work means the crate speeds up as it moves up the plane.

2.6 Link with energy (exam-useful extra)

DID YOU KNOW? **The work–energy principle**

Although a formal proof belongs to mechanics, it is very useful at GCE to know the **work–energy principle**: the total work done by all forces acting on a body equals the change in its kinetic energy:

$$W_{\text{total}} = \Delta\left(\tfrac{1}{2}mv^2\right) = \tfrac{1}{2}mv^2 - \tfrac{1}{2}mu^2.$$

In the previous example, the crate gained $\tfrac{1}{2}(20)(v^2 - u^2) = 217.3$ J of kinetic energy over the 8 m. This gives a fast way of finding speeds without resolving forces and using kinematics.

KEY POINTS **Key points**

- Work done by a constant force: $W = \vec{F} \cdot \vec{d} = |\vec{F}||\vec{d}| \cos\theta$, a scalar measured in joules.
- $\theta < 90^\circ$: positive work; $\theta = 90^\circ$: zero work; $\theta > 90^\circ$: negative work.
- With components, form $\vec{d} = \vec{AB}$ and take the dot product — no angle needed.
- Work done by gravity: $\pm mgh$, sign depending on whether the body falls or rises; only the vertical height matters.
- Total work = sum of individual works = work of the resultant. The normal reaction to a surface does no work.
- Work done against a resisting force is quoted as a positive number.

EXERCISE 2 **Practice**

1. A force $\vec{F} = (2\vec{i} - 3\vec{j} + \vec{k})$ N acts on a particle that moves from $A(1, 0, 2)$ to $B(4, 2, 5)$, distances in metres. Find the work done.
2. A girl drags a box 12 m across a horizontal floor with a strap inclined at 40° to the horizontal, exerting a force of 60 N. Find the work done by the pull.
3. A load of mass 50 kg is raised vertically through 3 m at constant speed. Find the work done by the lifting force and the work done by gravity. ($g = 9.8$ m s^{-2}.)
4. A trolley of mass 10 kg is pushed 5 m up a smooth slope inclined at 30° by a horizontal force. Using the work–energy principle, find the least work the pusher must supply if the trolley starts and ends at rest.

SOLUTION **Exercise 1**

1. $\vec{d} = \vec{AB} = 3\vec{i} + 2\vec{j} + 3\vec{k}$, so $W = (2)(3) + (-3)(2) + (1)(3) = 6 - 6 + 3 = 3$ J.
2. $W = 60 \times 12 \times \cos 40^\circ \approx 720 \times 0.766 = 551.5$ J.
3. Constant speed means the lifting force equals the weight, $T = mg = 490$ N. Work done by the lifting force: $W_T = 490 \times 3 = 1470$ J. Work done by gravity: $W_g = -mgh = -1470$ J (total work zero, consistent with constant speed).
4. Starting and ending at rest, $\Delta KE = 0$, so the pusher's work must exactly balance the work of gravity: $W = mgh = 10 \times 9.8 \times 5 \sin 30^\circ = 245$ J.

3 SECTION 3: MOMENT OF A FORCE AND EQUILIBRIUM (LESSON 104)

In the previous section we saw that the scalar product measures the work done by a force: it tells us how effectively a force moves a body along a straight line. But forces do not only translate bodies — they also turn them. When you push a door, tighten a nut with a spanner, or balance on a see-saw at the playground, you are producing a **turning effect**. Measuring that turning effect is the business of the vector product, and it leads us to one of the most powerful ideas in mechanics: the **moment of a force**.

3.1 The Turning Effect of a Force: Intuition First

EXPERIMENT **Three everyday observations**

- The door. Push a door near its hinges: it is very hard to open. Push it at the handle, far from the hinges: it opens easily. The same force has a greater turning effect when applied farther from the pivot.
- The spanner. A mechanic in Douala loosening a stubborn wheel nut uses a long spanner, and pulls perpendicularly to the handle. Pulling along the handle (towards the nut) produces no rotation at all.
- The see-saw. A small child can balance a heavy adult on a see-saw, provided the child sits far from the pivot and the adult sits close to it.

These observations tell us that the turning effect of a force about a point O depends on **two** things: the magnitude of the force, and the perpendicular distance from O to the line along which the force acts. This motivates our definition.

DEFINITION **Moment of a force about a point (2D)**

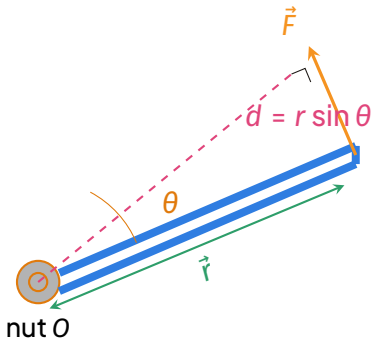
Let a force $\vec{F}$ act in a plane, and let O be a point of that plane. The **moment** of $\vec{F}$ about O is

$$M_O = F \times d,$$

where d is the perpendicular distance from O to the **line of action** of $\vec{F}$. The moment is a signed quantity: we attach + to one sense of rotation (usually anticlockwise) and − to the other. The SI unit of moment is the newton–metre, N m.

WARNING **Use the perpendicular distance!**

The distance d is not necessarily the distance from O to the point of application of the force. It is the shortest (perpendicular) distance from O to the line of action. In particular, the moment of a force about any point on its own line of action is zero, because $d = 0$. This is why pushing a door directly towards its hinges never opens it.



A spanner turning a nut: the moment about O is $M = Fd = Fr\sin\theta$.

3.2 The Vector Definition in Three Dimensions: $\vec{M} = \vec{r} \times \vec{F}$

In three dimensions the turning effect has not only a magnitude but also an axis of rotation. Both are captured elegantly by the vector product.

DEFINITION **Moment of a force about a point (3D)**

Let $\vec{F}$ be a force whose line of action passes through a point A, and let $\vec{r} = \vec{OA}$ be the position vector of A relative to O. The **moment** of $\vec{F}$ about O is the vector

$$\vec{M}_O = \vec{r} \times \vec{F}.$$

The result is unchanged if A is replaced by any other point of the line of action.

Why is the choice of point on the line of action irrelevant? If A' is another point of the line of action, then $\vec{OA'} = \vec{r} + \lambda\vec{F}$ for some scalar λ (since $\vec{AA'}$ is parallel to $\vec{F}$). Hence

$$\vec{OA'} \times \vec{F} = \vec{r} \times \vec{F} + \lambda(\vec{F} \times \vec{F}) = \vec{r} \times \vec{F},$$

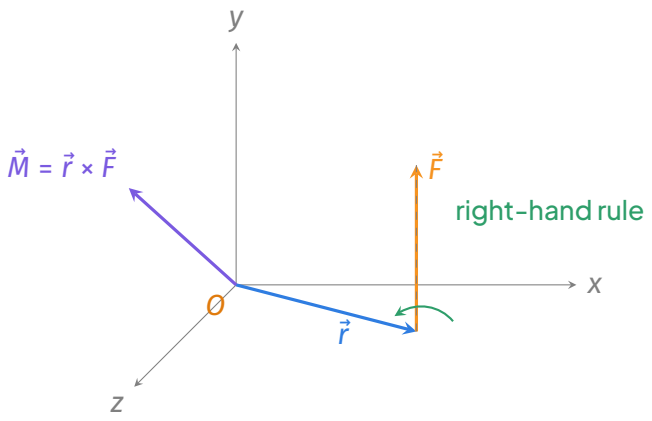
because the cross product of parallel vectors is zero.

PROPERTY **Magnitude and direction of the moment**

The magnitude of the moment is

$$|\vec{M}_O| = |\vec{r}|\,|\vec{F}| \sin\theta = F\,d,$$

where θ is the angle between $\vec{r}$ and $\vec{F}$, and $d = r\sin\theta$ is the perpendicular distance from O to the line of action. The direction of $\vec{M}_O$ is perpendicular to the plane containing $\vec{r}$ and $\vec{F}$, given by the **right-hand rule**: curl the fingers of the right hand from $\vec{r}$ towards $\vec{F}$; the thumb points along $\vec{M}_O$, i.e. along the axis about which the force tends to rotate the body.



In 3D, $\vec{M} = \vec{r} \times \vec{F}$ is perpendicular to the plane of $\vec{r}$ and $\vec{F}$; its sense is given by the right-hand rule.

EXEMPLE **Computing a moment in 3D**

A force $\vec{F} = (2\vec{i} + 3\vec{j} - \vec{k})$ N acts at the point $A(1, 2, 0)$. Find its moment about the origin O.

We have $\vec{r} = \vec{OA} = \vec{i} + 2\vec{j}$, so

$$\vec{M}_O = \vec{r} \times \vec{F} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \times \begin{pmatrix} 2 \\ 3 \\ -1 \end{pmatrix} = \begin{pmatrix} (2)(-1) - (0)(3) \\ (0)(2) - (1)(-1) \\ (1)(3) - (2)(2) \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \\ -1 \end{pmatrix} \text{N m.}$$

The magnitude is $|\vec{M}_O| = \sqrt{4 + 1 + 1} = \sqrt{6}$ N m.

3.3 Varignon's Theorem

PROPERTY **Varignon's theorem**

The moment about any point O of the resultant $\vec{R}$ of a system of forces $\vec{F}_1, \vec{F}_2, \ldots, \vec{F}_n$ is equal to the sum of the moments of the individual forces about the same point O:

$$\vec{M}_O(\vec{R}) = \sum_{i=1}^{n} \vec{M}_O(\vec{F}_i).$$

For a set of concurrent forces (all passing through a point A with position vector $\vec{r}$ relative to O), this reduces to $\vec{r} \times \vec{R} = \sum \vec{r} \times \vec{F}_i$, which follows immediately from the distributivity of the cross product over addition. A particularly useful corollary is that a single force may be replaced by its components acting at the same point when computing moments — often the fastest technique in 2D problems.

EXEMPLE **Using components (Varignon) in 2D**

A force of 10 N acts at the point $A(3, 4)$ in the direction of the vector $(1, 1)$. Find its moment about O.

The unit vector along $(1, 1)$ is $\frac{1}{\sqrt{2}}(1, 1)$, so $\vec{F} = \frac{10}{\sqrt{2}}(\vec{i} + \vec{j}) = 5\sqrt{2}\,\vec{i} + 5\sqrt{2}\,\vec{j}$. Taking moments of the components about O (with anticlockwise positive):

$$M_O = xF_y - yF_x = (3)(5\sqrt{2}) - (4)(5\sqrt{2}) = -5\sqrt{2} \text{ N m.}$$

The negative sign means the turning effect is clockwise, of magnitude $5\sqrt{2}$ N m.

3.4 Equilibrium of a Rigid Body and the Principle of Moments

A rigid body is in **equilibrium** when it neither translates nor rotates. Both possibilities must be blocked:

KEY POINTS **Conditions for equilibrium of a rigid body**

A rigid body is in equilibrium if and only if

1. the resultant force is zero: $\sum \vec{F} = \vec{0}$ (no translation);
2. the resultant moment about any point is zero: $\sum \vec{M}_O = \vec{0}$ (no rotation).

In 2D problems, condition 1 gives two scalar equations (resolving horizontally and vertically) and condition 2 gives one scalar equation (moments).

PROPERTY **Principle of moments**

When a body is in equilibrium, the sum of the clockwise moments about any point equals the sum of the anticlockwise moments about that same point:

$$\sum M_{\text{clockwise}} = \sum M_{\text{anticlockwise}}.$$

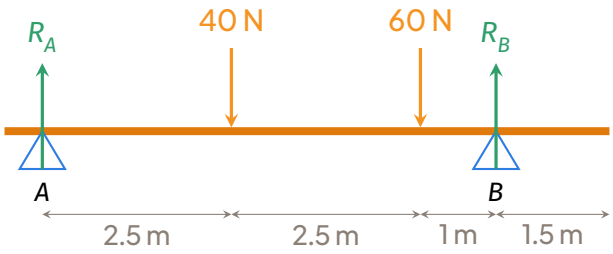
This is simply condition 2 restated, and it is the physical law behind every balance, lever and scale.

WARNING **Fix a sign convention before you start**

The most frequent source of error in GCE scripts is mixing signs: a candidate counts one clockwise moment as positive and another as negative in the same equation. Before writing any moment equation, decide: "anticlockwise is positive" (or the reverse), and apply it to every term. Alternatively, collect all clockwise moments on one side and all anticlockwise moments on the other.

METHOD **Solving beam and support problems**

1. Draw a clear diagram of the beam; mark all forces: loads, weights (acting at the centre of gravity of a uniform beam), and the unknown reactions at the supports.
2. Resolve forces vertically: $\sum F_y = 0$ (and horizontally if needed). This gives one equation linking the reactions.
3. Take moments about one of the supports: the reaction at that support then has zero moment (its line of action passes through the point), so it disappears from the equation, leaving an equation in the other reaction only.
4. Solve, then substitute back into the force equation to find the remaining reaction.
5. Check: take moments about the other support and verify consistency.



A beam on two supports A and B carrying two point loads. Take moments about A to find R_B directly.

WORKED EXAMPLE 6 **Reactions at the supports of a beam**

A light beam AC of length 7.5 m rests horizontally on supports at A and at B, where $AB = 6$ m and B is 1.5 m from the end C. A load of 40 N hangs 2.5 m from A, and a load of 60 N hangs 5 m from A. Find the reactions R_A and R_B.

SOLUTION

Step 1 — Resolve vertically. The beam is light (negligible weight), so

$$R_A + R_B = 40 + 60 = 100 \text{ N}. \tag{1}$$

Step 2 — Moments about A (anticlockwise positive; R_A has zero moment about A):

$$R_B \times 6 = 40 \times 2.5 + 60 \times 5 = 100 + 300 = 400,$$

$$R_B = \frac{400}{6} = \frac{200}{3} \approx 66.7 \text{ N}.$$

Step 3 — Back-substitute in (1):

$$R_A = 100 - \frac{200}{3} = \frac{100}{3} \approx 33.3 \text{ N}.$$

Check (moments about B): anticlockwise moment of R_A: $\frac{100}{3} \times 6 = 200$; clockwise moments: $40 \times 3.5 + 60 \times 1 = 140 + 60 = 200$. ✓ The two agree, so the beam is in equilibrium with $R_A = \frac{100}{3}$ N and $R_B = \frac{200}{3}$ N.

WORKED EXAMPLE 7 **The see-saw: principle of moments**

A uniform see-saw plank of length 4 m and weight 100 N is pivoted at its centre. A child of weight 250 N sits at one end. Where must a second child of weight 400 N sit so that the see-saw balances horizontally?

SOLUTION

The plank is uniform, so its weight acts at the pivot and has zero moment about it. Take moments about the pivot, anticlockwise positive, and let x be the distance of the second child from the pivot on the opposite side:

$$400\,x = 250 \times 2 \quad \Longrightarrow \quad x = \frac{500}{400} = 1.25\ \text{m}.$$

The second child must sit 1.25 m from the pivot, on the side opposite to the first child. Note how the heavier child sits closer to the pivot — exactly what everyday experience predicts.

DID YOU KNOW? **Moments in daily Cameroonian life**

The balance scale used by traders in the markets of Bamenda and Bafoussam is a direct application of the principle of moments: the unknown mass is placed on one pan and known masses on the other, and balance (zero resultant moment about the pivot) guarantees that the weights are equal. The same principle lets a carpenter use a long crowbar to lift a heavy stone with a modest effort: a small force with a large lever arm produces the same moment as a large force with a small lever arm. Archimedes summarised it long ago: "Give me a place to stand, and I shall move the Earth."

WARNING **Classic traps in moment problems**

- Forgetting the weight of a uniform beam: it acts at the midpoint of the beam, not at a support.
- Using the distance from O to the point of application instead of the perpendicular distance to the line of action when the force is oblique. When in doubt, resolve the force into components and use Varignon's theorem.
- Taking moments about a point and still including the moment of a force whose line of action passes through that point — its moment is zero.
- In 3D, computing $\vec{F} \times \vec{r}$ instead of $\vec{r} \times \vec{F}$: the order matters, since $\vec{F} \times \vec{r} = -\vec{r} \times \vec{F}$.

EXERCISE 3 **Ladder-type problem**

A uniform ladder of length 5 m and weight 200 N rests with its top against a smooth vertical wall and its foot on rough horizontal ground, making an angle of 60° with the ground.

The reaction of the smooth wall is horizontal; the ground exerts a vertical reaction and a friction force. By resolving forces and taking moments about the foot of the ladder, show that the wall reaction is $\dfrac{100}{\sqrt{3}}$ N ≈ 57.7 N, and find the friction force and the vertical ground reaction.

SOLUTION **Exercise 1**

Let N be the horizontal wall reaction, F the friction and R the vertical ground reaction. Resolving vertically: $R = 200$ N. Resolving horizontally: $F = N$. The weight acts at the midpoint, 2.5 m along the ladder; its perpendicular distance to the foot is $2.5\cos 60^\circ = 1.25$ m. The wall reaction acts at the top, at perpendicular distance $5\sin 60^\circ = \frac{5\sqrt{3}}{2}$ m from the foot. Moments about the foot (anticlockwise positive):

$$N \times \frac{5\sqrt{3}}{2} = 200 \times 1.25 = 250 \quad \Longrightarrow \quad N = \frac{500}{5\sqrt{3}} = \frac{100}{\sqrt{3}} \approx 57.7 \text{ N}.$$

Hence $F = \dfrac{100}{\sqrt{3}}$ N and $R = 200$ N.

KEY POINTS **Key points of Lesson 104**

- Moment of a force about O: $M = Fd$ in 2D (perpendicular distance), $\vec{M} = \vec{r} \times \vec{F}$ in 3D; unit N m.
- $|\vec{M}| = rF\sin\theta$; direction given by the right-hand rule; sense clockwise or anticlockwise in 2D.
- Varignon: the moment of the resultant equals the sum of the moments of the forces — resolve into components to simplify.
- Equilibrium requires both $\sum \vec{F} = \vec{0}$ and $\sum \vec{M}_O = \vec{0}$; in equilibrium, clockwise moments equal anticlockwise moments about any point.
- Strategy: resolve to zero, then take moments about a support to eliminate one unknown — and always fix a sign convention first.

With the moment of a single force mastered, we are ready for the next step: what happens when several forces act together — and in particular the remarkable pair of forces called a **couple**, which produces a pure turning effect with no translation at all. That is the subject of Section 4.

4 SECTION 4: SINGLE FORCES AND COUPLES (LESSON 105)

In Section 3 we studied the moment of a single force and saw that a body is in equilibrium when both the resultant force and the resultant moment vanish. But a natural question arises: what happens when the resultant force is zero yet the body still rotates? Anyone who has turned a steering wheel with both hands, or opened a water tap, already knows the answer. Two equal and opposite pushes applied at different points produce no translation at all, but a very definite rotation. This special pair of forces is called a **couple**, and it is the subject of this lesson. Understanding couples will then allow us to answer a deeper question: can any system of forces be reduced to something simple? The answer — a single force, or a force plus a couple — is one of the most powerful results in statics.

4.1 Definition of a Couple

EXPERIMENT **Turning the steering wheel**

Sit in a parked car (engine off, handbrake on — ask permission first!). Place your left hand at the top of the steering wheel and your right hand at the bottom. Push the top to the right and pull the bottom to the left with forces of equal magnitude. The centre of the wheel does not move: the two forces cancel as far as translation is concerned. Yet the wheel turns. The rotation is produced by the pair of forces, not by either force alone. The same effect occurs when you open a tap with two fingers, when a mechanic uses two spanners on a stubborn nut in a Douala workshop, or when you turn a screwdriver: the thumb pushes one way, the fingers push the other way.

DEFINITION **Couple**

A **couple** is a system of two forces $\vec{F}$ and $-\vec{F}$ which are:

- equal in magnitude,
- opposite in direction,
- parallel,
- and have different lines of action.

The plane containing the two lines of action is called the plane of the couple, and the perpendicular distance d between the two lines of action is called the arm of the couple.

WARNING **A couple is not a balanced system**

Two equal and opposite forces acting along the same line (like two teams pulling a rope with equal strength) cancel completely: no translation, no rotation. A couple is different precisely because the lines of action are distinct. Never confuse "equal and opposite forces" with "a couple": the separation $d \neq 0$ is essential.

4.2 Moment of a Couple

Let the force $\vec{F}$ act at point B and the force $-\vec{F}$ act at point A, and let O be any point. The total moment about O is

$$\vec{G}_O = \vec{OB} \wedge \vec{F} + \vec{OA} \wedge (-\vec{F}) = (\vec{OB} - \vec{OA}) \wedge \vec{F} = \vec{AB} \wedge \vec{F}.$$

The point O has disappeared from the result! This proves the fundamental property:

PROPERTY **Moment of a couple — proof examinable**

The moment of a couple is the same about every point. It is a **free vector**:

$$\vec{G} = \vec{r} \wedge \vec{F},$$

where $\vec{r}$ is the vector joining the point of application of $-\vec{F}$ to the point of application of $\vec{F}$. Its magnitude is

$$G = F\,d,$$

where F is the magnitude of either force and d is the perpendicular distance between the two lines of action (the arm). The SI unit is the newton metre, N m.

METHOD **Computing the moment of a couple**

1. Identify the two equal, opposite, parallel forces and their lines of action.
2. Measure (or compute) the perpendicular distance d between the two lines of action.
3. Multiply: $G = F \times d$.
4. Give the sense of rotation (clockwise or anticlockwise), or the vector direction by the right-hand rule.

WARNING **Use the perpendicular distance**

A very common mistake is to multiply F by the distance AB between the points of appli-

cation instead of the perpendicular distance between the lines of action. If $\vec{AB}$ makes an angle θ with $\vec{F}$, then $d = AB\sin\theta$ and $G = F \cdot AB\sin\theta$ — exactly what the cross product $\vec{AB} \wedge \vec{F}$ gives automatically.

WORKED EXAMPLE 8 **Moment of a couple on a steering wheel**

A driver applies two forces of magnitude $F = 15$ N tangentially at opposite ends of a diameter of a steering wheel of radius 20 cm. Find the moment of the couple.

SOLUTION

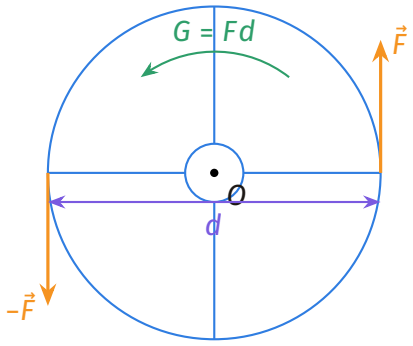
Solution. The two forces are equal, opposite and parallel, with lines of action separated by the diameter:

$$d = 2 \times 0.20 = 0.40 \text{ m}.$$

Hence

$$G = F\,d = 15 \times 0.40 = 6 \text{ N m}.$$

The moment is 6 N m about any point — the centre of the wheel, the rim, or even a point outside the car.



A couple applied to a steering wheel: two equal opposite tangential forces at the rim, arm d equal to the diameter, moment $G = Fd$.

4.3 A Couple Produces Pure Rotation

The vector sum of the two forces of a couple is

$$\vec{F} + (-\vec{F}) = \vec{0},$$

so a couple has **zero resultant force**: it cannot accelerate the centre of mass of a body. But its resultant moment $G = Fd \neq 0$, so it produces **pure rotation**. A couple is the purest form of "turning effect": it is to rotation what a single force is to translation.

WARNING **A couple cannot be balanced by a single force**

Since the resultant force of a couple is zero, no single force can ever cancel its turning effect. A couple can only be balanced by another couple of equal magnitude and opposite sense. In equilibrium problems, if the applied loads include a couple of moment G, the supports must provide a resisting couple $-G$; a reaction force alone will not do.

4.4 Equivalence of Couples

Because the moment of a couple depends only on the product Fd (and the sense of rotation), two couples are **equivalent** whenever they have the same moment vector, whatever the individual forces and arms.

PROPERTY **Equivalence of couples**

Two couples are equivalent if and only if their moment vectors are equal. Consequently a couple may be replaced by any other couple with the same moment: for example, forces of 10 N with arm 0.6 m are equivalent to forces of 30 N with arm 0.2 m, since both give $G = 6\,\text{N m}$.

EXEMPLE **Two spanners on a nut**

A mechanic loosens a nut using two spanners, pushing with forces of 40 N at a separation of 25 cm: $G = 40 \times 0.25 = 10\,\text{N m}$. His apprentice, with shorter spanners separated by 12.5 cm, must push with forces of $\dfrac{10}{0.125} = 80\,\text{N}$ to produce the same turning effect. Same couple, different forces.

4.5 Translating a Force: Force Plus a Couple

Here is a beautiful trick. Let a force $\vec{F}$ act at a point A, and let O be any other point. Add at O two forces $\vec{F}$ and $-\vec{F}$ (their sum is zero, so nothing has changed). Now look at what we have:

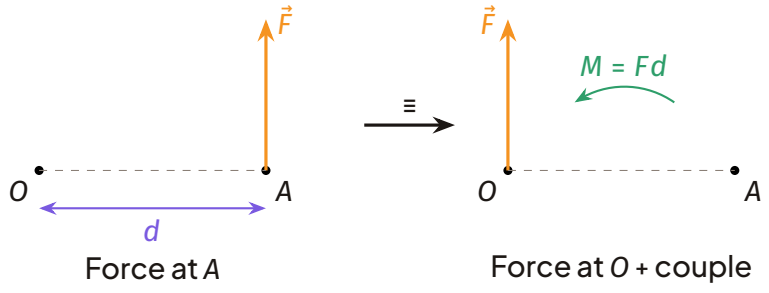
- the original $\vec{F}$ at A together with $-\vec{F}$ at O form a **couple** of moment $\vec{OA} \wedge \vec{F}$;
- and there remains a force $\vec{F}$ acting at O.

PROPERTY **Force translation theorem — proof examinable**

A force $\vec{F}$ acting at a point A is equivalent to an equal parallel force $\vec{F}$ acting at any other point O, together with a couple of moment

$$\vec{M} = \vec{OA} \wedge \vec{F}, \qquad M = Fd,$$

where d is the perpendicular distance from O to the original line of action.



Force translation theorem: $\vec{F}$ at A is equivalent to the same force at O plus a couple of moment $M = Fd$.

WORKED EXAMPLE 9 **Translating a force**

A force $\vec{F} = 50$ N acts vertically upwards at point A, where $\vec{OA} = 0.3$ m horizontally to the right of O. Replace it by a force at O plus a couple.

SOLUTION

Solution. By the force translation theorem, we place a force of 50 N vertically upwards at O, together with a couple of moment

$$M = Fd = 50 \times 0.3 = 15 \text{ N m}.$$

The sense of the couple is anticlockwise (the original force tends to rotate the body about O anticlockwise). Check with the cross product: with $\vec{OA} = 0.3\,\vec{i}$ and $\vec{F} = 50\,\vec{j}$,

$$\vec{M} = \vec{OA} \wedge \vec{F} = 0.3 \times 50\,(\vec{i} \wedge \vec{j}) = 15\,\vec{k} \text{ N m},$$

confirming the anticlockwise sense about O.

4.6 Reduction of a System of Forces at a Point

The force translation theorem gives us a complete recipe for simplifying any system of forces. Take a system of forces $\vec{F}_1, \vec{F}_2, \ldots, \vec{F}_n$ acting at points $A_1, A_2, \ldots, A_n$, and choose a point O (the centre of reduction). Translate each force to O: each $\vec{F}_i$ contributes a force $\vec{F}_i$ at O plus a couple $\vec{OA}_i \wedge \vec{F}_i$. Adding everything:

PROPERTY **Reduction of a force system**

Any system of forces is equivalent, at an arbitrary point O, to

- a single resultant force $\vec{R} = \sum_i \vec{F}_i$ acting through O,
- together with a single resultant couple $\vec{G}_O = \sum_i \vec{OA}_i \wedge \vec{F}_i$.

Special cases:

- If $\vec{R} \neq \vec{0}$ and $\vec{G}_O$ is perpendicular to $\vec{R}$ (in particular for coplanar forces), the system can be reduced further to a single force $\vec{R}$ alone, acting along a line at perpendicular distance $d = \dfrac{G_O}{R}$ from O.
- If $\vec{R} = \vec{0}$ and $\vec{G}_O \neq \vec{0}$, the system reduces to a couple.
- If $\vec{R} = \vec{0}$ and $\vec{G}_O = \vec{0}$, the system is in equilibrium.
- In general three-dimensional systems, the reduction yields a **wrench**: a force $\vec{R}$ together with a couple $\vec{G}_{\parallel}$ parallel to $\vec{R}$ (the component of $\vec{G}_O$ along $\vec{R}$ cannot be eliminated). This is studied in Lesson 107.

WORKED EXAMPLE 10 **Reducing a system of two parallel forces**

Two parallel forces act on a horizontal bar: $\vec{F}_1 = 20$ N downwards at A_1 with $OA_1 = 0.2$ m, and $\vec{F}_2 = 50$ N downwards at A_2 with $OA_2 = 0.8$ m. Reduce the system at O, then find the single force to which it is equivalent.

SOLUTION

Solution. Step 1: resultant force.

$$\vec{R} = \vec{F}_1 + \vec{F}_2 = (20 + 50) = 70 \text{ N downwards, acting at } O.$$

Step 2: resultant couple at O (taking anticlockwise as positive, moments of downward forces to the right of O are clockwise):

$$G_O = -(20 \times 0.2) - (50 \times 0.8) = -4 - 40 = -44 \text{ N m},$$

i.e. 44 N m clockwise. Step 3: single force. Since $\vec{R} \neq \vec{0}$ and the forces are coplanar, the system reduces to a single force of 70 N downwards whose line of action is at distance

$$d = \frac{G_O}{R} = \frac{44}{70} \approx 0.629 \text{ m}$$

to the right of O. Check: $70 \times 0.629 \approx 44$ N m clockwise. ✓

EXERCISE 4 **Couples and reduction**

1. A couple consists of two forces of 25 N with arm 0.36 m. Find its moment. What forces would give an equivalent couple with arm 0.60 m?
2. A force of 80 N acts horizontally at A, with O at perpendicular distance 0.45 m from its line of action. Replace it by a force at O plus a couple.
3. Two forces of 30 N downwards act at distances 0.1 m and 0.5 m from O on a bar. Reduce the system to a single force and give the position of its line of action.

SOLUTION **Exercise 1**

1. $G = Fd = 25 \times 0.36 = 9$ N m. For an equivalent couple with arm 0.60 m: $F = \dfrac{9}{0.60} = 15$ N.
2. Force of 80 N at O (same direction) plus a couple $M = Fd = 80 \times 0.45 = 36$ N m, in the sense in which the original force turns about O.
3. $\vec{R} = 60$ N downwards; $G_O = 30 \times 0.1 + 30 \times 0.5 = 18$ N m clockwise. Single force of 60 N at distance $d = \dfrac{18}{60} = 0.3$ m from O (the midpoint, as symmetry suggests).

KEY POINTS **Key points of Lesson 105**

- A **couple** is a pair of equal, opposite, parallel forces with distinct lines of action.
- Its moment $\vec{G} = \vec{r} \wedge \vec{F}$, of magnitude $G = Fd$, is a **free vector**: the same about every point.
- A couple has zero resultant force: it produces **pure rotation** and can only be balanced by another couple.
- Couples with the same moment are **equivalent**.
- A force at A is equivalent to the same force at O plus a couple $\vec{OA} \wedge \vec{F}$ (**force translation theorem**).
- Any force system reduces, at a point O, to a resultant force $\vec{R}$ plus a resultant couple $\vec{G}_O$; if $\vec{R} \neq \vec{0}$ and $\vec{G}_O \perp \vec{R}$ (coplanar case) it reduces further to a single force at distance G_O / R from O.

DID YOU KNOW? **Couples all around us**

Every rotating machine obeys the physics of couples. The electric motors that power the fans in our homes deliver a couple (called torque) to the shaft; the pedals of a bicycle work best when the rider pushes with two feet in opposition, forming a couple on the crank; and when carpenters in Bamenda tighten a vice, they instinctively push the two ends of the handle in opposite directions — applying a couple rather than a single off-centre force, which would bend the screw. Engineers design steering systems, taps and door handles precisely so that our two hands naturally form couples.

5 SECTION 5: COPLANAR FORCES AND SIMPLE 3D SYSTEMS (LESSONS 106 & 107)

In Section 4 we saw that a system of forces acting on a rigid body can often be reduced either to a **single force** (its resultant) or to a **couple** (a pure turning effect with zero resultant force). We now put everything together. In this final section we answer three big questions that the GCE examiner loves: What is the resultant of a whole system of coplanar forces, and where does it act? When is a body in equilibrium under such a system? And what changes when the forces live in three dimensions, as with a particle hanging from cables? Mastering this section is mastering the whole chapter.

5.1 Resultant of a System of Coplanar Forces

Idea. Suppose several forces $\vec{F}_1, \vec{F}_2, \dots, \vec{F}_n$ act on a rigid body in the same plane (the xOy plane). The **resultant** $\vec{R}$ is simply their vector sum, computed component by component:

$$\vec{R} = \sum_{i=1}^{n} \vec{F}_i, \qquad R_x = \sum F_{ix}, \qquad R_y = \sum F_{iy}.$$

Its magnitude and direction follow at once:

$$|\vec{R}| = \sqrt{R_x^2 + R_y^2}, \qquad \tan\theta = \frac{R_y}{R_x},$$

where θ is the angle the resultant makes with the positive x-axis (take care to place θ in the correct quadrant from the signs of R_x and R_y).

Where does the resultant act? Knowing $\vec{R}$ is not enough: a force applied at different points of a rigid body produces different turning effects. The **line of action** of the resultant is located by taking moments. If the system is equivalent to the single force $\vec{R}$, then the moment of $\vec{R}$ about any point O must equal the sum of the moments of the individual forces about O:

$$\vec{R} \text{ acting along its line of action} \iff M_O(\vec{R}) = \sum_{i=1}^{n} M_O(\vec{F}_i).$$

Recall from Section 3 that for a force (F_x, F_y) applied at the point (x, y), the moment about the origin is $M_O = xF_y - yF_x$. In practice, if the line of action of $\vec{R}$ cuts the x-axis at $(\bar{x}, 0)$, then $M_O(\vec{R}) = \bar{x}\,R_y$ (the component R_x passes through the axis and gives no moment), so

$$\bar{x} = \frac{\sum M_O(\vec{F}_i)}{R_y}.$$

If $R_y = 0$ but $R_x \neq 0$, use instead the point $(0, \bar{y})$ where the line of action cuts the y-axis: $M_O(\vec{R}) = -\bar{y}\,R_x$, so $\bar{y} = -\dfrac{\sum M_O(\vec{F}_i)}{R_x}$.

METHOD **Reducing a coplanar system to a single resultant**

1. Resolve every force into x and y components; sum to get R_x and R_y.
2. Compute $|\vec{R}| = \sqrt{R_x^2 + R_y^2}$ and $\theta = \tan^{-1}(R_y / R_x)$ in the correct quadrant.
3. Take moments of all the original forces about a convenient point O (usually the origin), using $M_O = xF_y - yF_x$.
4. Locate the line of action from $\bar{x} = \dfrac{\sum M_O}{R_y}$ (or, if $R_y = 0$, use $\bar{y} = -\dfrac{\sum M_O}{R_x}$ on the y-axis).
5. If $\vec{R} = \vec{0}$ but $\sum M_O \neq 0$, the system reduces to a couple of moment $\sum M_O$.

WORKED EXAMPLE 11 **Resultant and its line of action**

Forces act on a lamina in the xOy plane: $\vec{F}_1 = (3\vec{i} + 4\vec{j})$ N at $A(0, 0)$, $\vec{F}_2 = (-5\vec{i} + 2\vec{j})$ N at $B(4, 0)$, and $\vec{F}_3 = (4\vec{i} - 6\vec{j})$ N at $C(2, 3)$. Find the resultant and the equation of its line of action.

SOLUTION

Solution. Components of the resultant:

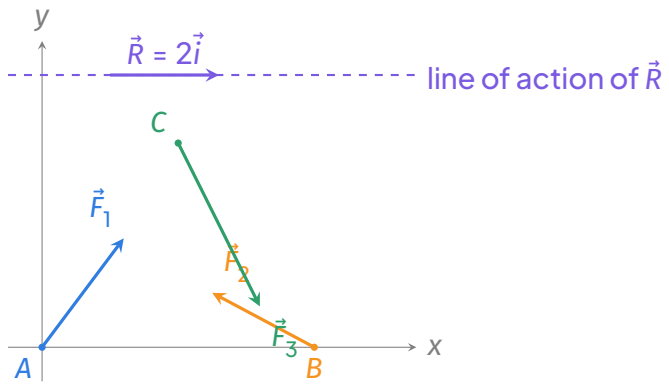
$$R_x = 3 - 5 + 4 = 2\ \text{N}, \qquad R_y = 4 + 2 - 6 = 0\ \text{N}.$$

So $\vec{R} = 2\vec{i}$ N: a horizontal force of 2 N. Since $R_y = 0$, we locate the line of action using the y-intercept instead. Moments about O (using $M_O = xF_y - yF_x$):

$$M_O(\vec{F}_1) = 0 \quad \text{(acts at } O\text{)},$$
$$M_O(\vec{F}_2) = 4(2) - 0(-5) = 8\ \text{N m},$$
$$M_O(\vec{F}_3) = 2(-6) - 3(4) = -12 - 12 = -24\ \text{N m}.$$

Total: $\sum M_O = 8 - 24 = -16$ N m. If $\vec{R} = 2\vec{i}$ acts at $(0, \bar{y})$, its moment about O is $-\bar{y}R_x = -2\bar{y}$.

Hence $-2\bar{y} = -16$, giving $\bar{y} = 8$ m. The resultant is the force $2\vec{i}$ N acting along the horizontal line $y = 8$.



A coplanar system reduced to a single resultant: the three forces are equivalent to $\vec{R} = 2\vec{i}$ N acting along a horizontal line (schematic; the computation gives $y = 8$).

5.2 Equilibrium of a Body Under Coplanar Forces

A rigid body is in **equilibrium** when it neither translates nor rotates. Translating is prevented by requiring the resultant force to vanish; rotating is prevented by requiring the total moment to vanish.

PROPERTY **Conditions of equilibrium under coplanar forces (proof examinable in outline)**

A rigid body is in equilibrium under a system of coplanar forces if and only if

$$\sum F_x = 0, \qquad \sum F_y = 0, \qquad \sum M_O = 0 \quad \text{for any point } O.$$

Justification. If $\sum F_x = \sum F_y = 0$, the resultant force is zero, so the system reduces (by Section 4) either to nothing or to a couple. The couple, if any, has the same moment $\sum M_O$ about every point; imposing $\sum M_O = 0$ kills the couple too. Nothing remains: the body is in equilibrium. Conversely, if the body is in equilibrium, it has no tendency to translate (so $\sum F_x = \sum F_y = 0$) and no tendency to rotate about any point (so $\sum M_O = 0$).

WARNING **Only three independent equations**

In the plane you have exactly three independent equilibrium equations, so you can deter-

mine at most three unknowns. Writing moments about a second point gives an equation that is a consequence of the first three — useful as a check, not as extra information.

METHOD **Choosing the moment centre wisely**

Take moments about the point of intersection of two unknown forces. Both unknowns then have zero moment arm and disappear from the moment equation, leaving an equation in the third unknown alone. This is the single most powerful time-saver in GCE statics problems.

WORKED EXAMPLE 12 **Ladder against a wall**

A uniform ladder AB of length 5 m and weight $W = 200$ N leans against a smooth vertical wall at A and rests on rough horizontal ground at B, with AB making 60° with the ground. Find the reaction at the wall and the frictional force at the ground.

SOLUTION

Solution. The wall is smooth, so its reaction R_A is horizontal (perpendicular to the wall); the ground reaction has a vertical component N (the normal reaction) and a horizontal component F (friction). Forces on the ladder: W at the midpoint, R_A at A, N and F at B.

$$\sum F_y = 0: \quad N = W = 200\ \text{N}; \qquad \sum F_x = 0: \quad F = R_A.$$

Take moments about B (this eliminates the two unknowns N and F acting at B). Perpendicular distances from B: the line of action of W passes through the midpoint, at horizontal distance $\frac{5}{2}\cos 60^\circ = 1.25$ m from B; the line of action of R_A passes through A, at vertical height $5\sin 60^\circ = \frac{5\sqrt{3}}{2}$ m above B. The weight tends to turn the ladder clockwise about B, the wall reaction anticlockwise:

$$\sum M_B = 0: \quad R_A \cdot \frac{5\sqrt{3}}{2} - 200 \times 1.25 = 0 \implies R_A = \frac{250}{4.330} \approx 57.7\ \text{N}.$$

Hence $F = R_A \approx 57.7$ N and $N = 200$ N.

5.3 Three Coplanar Forces in Equilibrium

PROPERTY **Concurrency of three forces in equilibrium**

If a rigid body is in equilibrium under exactly three non-parallel coplanar forces, then the

three forces are **concurrent** (their lines of action meet at one point).

Proof. Suppose two of the forces, $\vec{F}_1$ and $\vec{F}_2$, meet at a point P (they are not parallel, so they must meet). Take moments about P: $\vec{F}_1$ and $\vec{F}_2$ contribute nothing, since their lines of action pass through P. Equilibrium requires $\sum M_P = 0$, so the moment of $\vec{F}_3$ about P is zero, which means the line of action of $\vec{F}_3$ also passes through P. ■

Consequently, since $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$, the three force vectors placed head to tail form a closed triangle — the **triangle of forces** — and from the sine rule in that triangle we obtain a flagged, exam-useful extra:

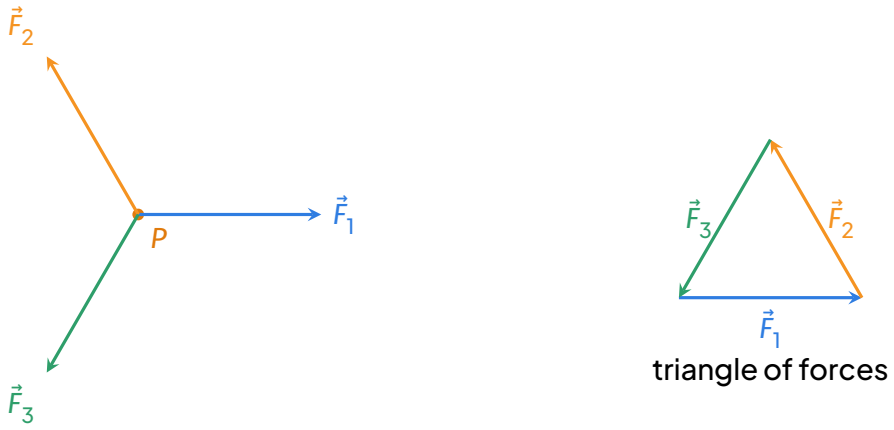
PROPERTY **Lami's theorem (useful extra, not a substitute for resolution)**

If three concurrent forces $\vec{P}, \vec{Q}, \vec{R}$ keep a body in equilibrium, and α, β, γ are the angles opposite to them (i.e. the angles between the other two forces), then

$$\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}.$$

WARNING **When Lami's theorem may NOT be used**

Do not apply Lami's theorem unless exactly three forces act and they are concurrent. A ladder on rough ground has four forces (weight, two reaction components at the ground, wall reaction): Lami is forbidden there. When in doubt, resolve and take moments — that method always works.



Three concurrent coplanar forces in equilibrium (left); placed head to tail they form a closed triangle (right), from which Lami's theorem follows by the sine rule.

EXEMPLE **Reduction to a couple**

Forces 5 N act along AB and 5 N along CD of a square $ABCD$ of side 2 m, in opposite senses, with $AB \parallel CD$. Then $\sum \vec{F} = \vec{0}$, but moments about the centre give $5 \times 1 + 5 \times 1 = 10$ N m (each force has moment arm 1 m from the centre, and both turn the same way). The system cannot be reduced to a single force: it is a couple of moment 10 N m. Remember the test: $\vec{R} = \vec{0}$ and $\sum M \neq 0 \Rightarrow$ couple.

5.4 Simple Systems of Forces in Three Dimensions

Many real structures — a mast held by cables on a building site in Douala, a sign hanging from three ropes — are genuinely three-dimensional. Fortunately, the ideas extend naturally.

Resultant of concurrent forces in 3D. If forces $\vec{F}_1, \dots, \vec{F}_n$ act at the same particle, their resultant is

$$\vec{R} = \sum \vec{F}_i = \Big(\sum F_{ix}\Big)\vec{i} + \Big(\sum F_{iy}\Big)\vec{j} + \Big(\sum F_{iz}\Big)\vec{k}, \qquad |\vec{R}| = \sqrt{R_x^2 + R_y^2 + R_z^2}.$$

Equilibrium of a particle in 3D. A particle is in equilibrium if and only if $\sum \vec{F} = \vec{0}$, i.e. the **three** scalar equations

$$\sum F_x = 0, \qquad \sum F_y = 0, \qquad \sum F_z = 0.$$

METHOD **The unit-vector method for cable problems**

1. Write the position vector of each anchor point relative to the particle, e.g. $\vec{PA} = \vec{OA} - \vec{OP}$.
2. Compute its length $|\vec{PA}|$ and form the unit vector $\hat{u} = \dfrac{\vec{PA}}{|\vec{PA}|}$ along the cable.
3. Write each cable force as $\vec{T} = T\,\hat{u}$ (tension times unit vector).
4. Sum all forces, set each of the three components to zero, and solve the resulting system.

WARNING **Three equations, not two**

In 3D, equilibrium requires all three component equations. Solving only $\sum F_x = 0$ and $\sum F_y = 0$ and forgetting $\sum F_z = 0$ is a frequent and costly error — especially when a weight acts in the z-direction.

WORKED EXAMPLE 13 **Particle held by three cables**

A particle P of weight 100 N hangs at $P(1, 1, 0)$, held by three cables anchored at $A(0, 0, 2)$,

$B(2,0,2)$ and $C(1,3,2)$. Find the tensions T_1, T_2, T_3 in PA, PB, PC.

SOLUTION

Solution. Position vectors from P:

$$\vec{PA} = -\vec{i} - \vec{j} + 2\vec{k}, \quad \vec{PB} = \vec{i} - \vec{j} + 2\vec{k}, \quad \vec{PC} = 0\vec{i} + 2\vec{j} + 2\vec{k},$$

with lengths $|\vec{PA}| = |\vec{PB}| = \sqrt{1+1+4} = \sqrt{6}$ and $|\vec{PC}| = \sqrt{0+4+4} = \sqrt{8} = 2\sqrt{2}$. The forces on P are

$$\frac{T_1}{\sqrt{6}}(-\vec{i} - \vec{j} + 2\vec{k}), \quad \frac{T_2}{\sqrt{6}}(\vec{i} - \vec{j} + 2\vec{k}), \quad \frac{T_3}{2\sqrt{2}}(2\vec{j} + 2\vec{k}), \quad -100\vec{k}.$$

Component equations:

$$x: \quad \frac{-T_1 + T_2}{\sqrt{6}} = 0 \Rightarrow T_1 = T_2,$$

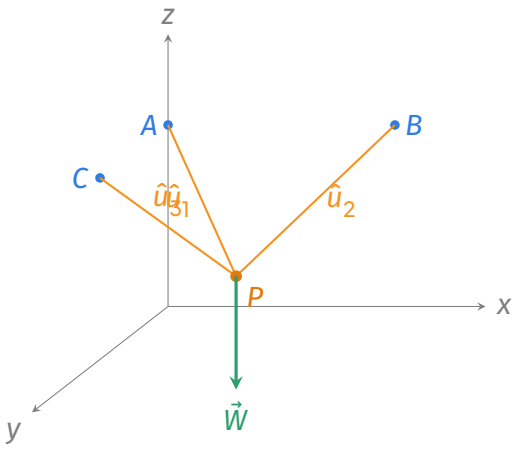
$$y: \quad \frac{-T_1 - T_2}{\sqrt{6}} + \frac{T_3}{\sqrt{2}} = 0,$$

$$z: \quad \frac{2T_1 + 2T_2}{\sqrt{6}} + \frac{T_3}{\sqrt{2}} - 100 = 0.$$

Setting $T_1 = T_2 = T$: the y-equation gives $\frac{-2T}{\sqrt{6}} + \frac{T_3}{\sqrt{2}} = 0$, so $T_3 = \frac{2T\sqrt{2}}{\sqrt{6}} = \frac{2T}{\sqrt{3}}$. Then $\frac{T_3}{\sqrt{2}} = \frac{2T}{\sqrt{6}}$, and substituting in the z-equation:

$$\frac{4T}{\sqrt{6}} + \frac{2T}{\sqrt{6}} = 100 \Rightarrow \frac{6T}{\sqrt{6}} = 100 \Rightarrow T = \frac{100\sqrt{6}}{6} \approx 40.8\,\text{N}.$$

Hence $T_1 = T_2 \approx 40.8$ N and $T_3 = \frac{2 \times 40.8}{\sqrt{3}} \approx 47.1$ N.



A particle P held by three cables anchored at A, B, C. Each tension acts along the unit vector from P towards its anchor; the weight acts vertically downwards.

Moments in 3D (extension for strong candidates). The moment of a force $\vec{F}$ about a point O is the vector

$$\vec{M}_O = \vec{r} \times \vec{F},$$

where $\vec{r}$ is the position vector of any point on the line of action of $\vec{F}$. A rigid body is in equilibrium in 3D when $\sum \vec{F} = \vec{0}$ and $\sum \vec{M}_O = \vec{0}$ — six scalar equations in all. Only simple cases (e.g. a single moment computed with the cross product) are expected at this level.

EXEMPLE **Moment via the cross product**

A force $\vec{F} = (2\vec{i} + \vec{j} - \vec{k})$ N acts at the point $(1, 0, 2)$. Its moment about O is

$$\vec{M}_O = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0(-1) - 2(1) \\ 2(2) - 1(-1) \\ 1(1) - 0(2) \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix} \text{ N m}.$$

Check: the y-component is $r_z F_x - r_x F_z = 2(2) - 1(-1) = 5$, as required by the cross-product formula.

5.5 Synthesis: GCE-Style Combined Problems

Examination questions typically combine resolution, moments and (sometimes) couples in one situation. The strategy never changes: draw a clear force diagram, resolve, and take moments about the intersection of two unknowns.

WORKED EXAMPLE 14 **Hinged rod with a cable**

A uniform rod AB of length 4 m and weight 80 N is hinged at A to a wall and held horizontal by a cable from B to a point C on the wall 3 m above A. Find the tension in the cable and the horizontal and vertical components of the reaction at A.

SOLUTION

Solution. The cable BC has length $\sqrt{4^2+3^2}=5$ m, so the tension T at B pulls along BC with components $\left(-\frac{4}{5}T,\ \frac{3}{5}T\right)$. Let the hinge reaction be (H, V). Moments about A (eliminating H and V): the weight acts at the midpoint, 2 m from A; the horizontal part of T acts along the rod and has no moment about A, so only the vertical part of T contributes:

$$\sum M_A = 0: \quad \frac{3}{5}T \times 4 - 80 \times 2 = 0 \ \Rightarrow\ T = \frac{160 \times 5}{12} \approx 66.7\ \text{N}.$$

Then

$$\sum F_x = 0:\ H = \frac{4}{5}T \approx 53.3\ \text{N}; \qquad \sum F_y = 0:\ V = 80 - \frac{3}{5}T = 80 - 40 = 40\ \text{N}.$$

EXERCISE 5 **Practice**

1. Forces $(2\vec{i}+\vec{j})$ N at $(1,0)$, $(-\vec{i}+3\vec{j})$ N at $(0,2)$ and $(-\vec{i}-4\vec{j})$ N at $(3,1)$ act on a lamina. Show that the system reduces to a couple and find its moment.
2. A particle of weight 50 N hangs at $P(0,0,0)$ from two cables anchored at $A(0,3,4)$ and $B(0,-3,4)$. Find the two tensions.
3. Three concurrent forces keep a body in equilibrium; two of them are 10 N and 14 N with an angle of 120° between them. Use the triangle of forces to find the third force.

SOLUTION **Exercise 1**

$R_x = 2-1-1 = 0$ and $R_y = 1+3-4 = 0$, so $\vec{R} = \vec{0}$. Moments about O (using $M_O = xF_y - yF_x$): $M_1 = 1(1) - 0(2) = 1$; $M_2 = 0(3) - 2(-1) = 2$; $M_3 = 3(-4) - 1(-1) = -12+1 = -11$. Total $\sum M_O = 1+2-11 = -8$ N m $\neq 0$: the system is a couple of moment 8 N m clockwise.

SOLUTION **Exercise 2**

$\vec{PA} = (0,3,4)$ and $\vec{PB} = (0,-3,4)$, with $|\vec{PA}| = |\vec{PB}| = \sqrt{9+16} = 5$. The forces on P are $\frac{T_1}{5}(3\vec{j}+4\vec{k})$,

$\frac{T_2}{5}(-3\vec{j} + 4\vec{k})$ and $-50\vec{k}$.

$$\sum F_y = 0 : \frac{3T_1 - 3T_2}{5} = 0 \Rightarrow T_1 = T_2; \qquad \sum F_z = 0 : \frac{4T_1 + 4T_2}{5} = 50.$$

With $T_1 = T_2 = T$: $\frac{8T}{5} = 50$, so $T = \frac{250}{8} = 31.25$ N. Hence $T_1 = T_2 = 31.25$ N. (The x-equation is satisfied identically: the geometry has no x-components.)

SOLUTION **Exercise 3**

In the triangle of forces, the angle between the sides representing the 10 N and 14 N forces is the supplement $180^\circ - 120^\circ = 60^\circ$ (head-to-tail placement reverses one force). By the cosine rule, the third force R closes the triangle:

$$R^2 = 10^2 + 14^2 - 2(10)(14)\cos 60^\circ = 100 + 196 - 140 = 156,$$

so $R = \sqrt{156} \approx 12.5$ N.

KEY POINTS **Key points of Section 5**

- Resultant of coplanar forces: $\vec{R} = \sum \vec{F}_i$ by components; its line of action is fixed by $\sum M_O(\vec{F}_i) = M_O(\vec{R})$, with $M_O = xF_y - yF_x$.
- Equilibrium in the plane: $\sum F_x = 0, \sum F_y = 0, \sum M_O = 0$ — three equations, three unknowns at most.
- Take moments about the intersection of two unknown forces to eliminate them.
- Three non-parallel coplanar forces in equilibrium are concurrent; then (and only then) the triangle of forces and Lami's theorem apply.
- $\vec{R} = \vec{0}$ with $\sum M \neq 0$ means the system is a couple.
- In 3D: write each cable force as $T\,\hat{u}$ with $\hat{u} = \vec{PA}/|\vec{PA}|$, and solve all three component equations; moments use $\vec{M}_O = \vec{r} \times \vec{F}$.

6 INTEGRATION ACTIVITY

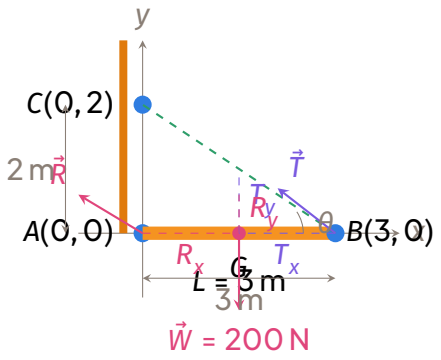
7 COMPREHENSIVE APPLICATION: STATIC ANALYSIS OF THE MILE 2 MARKET AWNING

7.1 Aims of the Activity

- Consolidate the concepts of vector components, work done by a constant force, moment of a force, couples, and equilibrium of coplanar forces in a single authentic engineering task.
- Develop the ability to model a real-world structure using rigid-body mechanics: drawing a free-body diagram, resolving forces, applying equilibrium conditions ($\sum \vec{F} = \vec{0}, \sum \vec{M} = \vec{0}$), and interpreting the physical meaning of the results.
- Practise systematic problem-solving and clear technical communication.

7.2 Problem Statement

In the bustling Mile 2 Market in Bamenda, a trader named Mama Ngo wants to install a sturdy horizontal awning to protect her goods. The awning is modelled as a **uniform plank** of weight $W = 200$ N and length $L = 3$ m. One end A is hinged to a vertical wall; the other end B is supported by a light inextensible cable attached to a point C on the wall, vertically 2 m above the hinge A. The cable makes an angle θ with the horizontal. The system is in static equilibrium.



Free-body diagram of the awning: hinge at A, cable BC, weight $\vec{W}$ at centre G, tension $\vec{T}$, reaction $\vec{R}$. Components shown dashed.

The trader asks a group of Upper Sixth Further Mathematics students to carry out a complete static analysis. The analysis must determine:

1. The angle θ and the vector components of all forces.
2. The tension T in the cable using moment equilibrium about A.

3. The reaction $\vec{R}$ at the hinge using force equilibrium.
4. The work done by the tension if the awning is lowered slightly ($\delta y = 0.01$ m at B).
5. The effect of adding a second symmetric cable from B to $D(0, -2)$: couple formation, concurrency condition, and viability.
6. A safety assessment (cable max 500 N, hinge max 400 N).

7.3 Guided Solution

1. Geometry and Vector Components

The cable runs from $B(3, 0)$ to $C(0, 2)$. The horizontal distance is 3 m, the vertical distance is 2 m.

$$\tan\theta = \frac{2}{3} \quad \Rightarrow \quad \theta = \arctan\left(\frac{2}{3}\right) \approx 33.69^\circ.$$

The length of the cable is $|\vec{BC}| = \sqrt{3^2 + 2^2} = \sqrt{13}$ m. The unit vector along $\vec{BC}$ (from B toward C) is

$$\hat{u}_{BC} = \frac{\vec{BC}}{|\vec{BC}|} = \frac{-3\vec{i} + 2\vec{j}}{\sqrt{13}}.$$

Since the tension pulls B toward C, the tension vector is

$$\vec{T} = T\,\hat{u}_{BC} = T\left(-\frac{3}{\sqrt{13}}\vec{i} + \frac{2}{\sqrt{13}}\vec{j}\right) = -\frac{3T}{\sqrt{13}}\vec{i} + \frac{2T}{\sqrt{13}}\vec{j}.$$

Thus the components are $T_x = -\dfrac{3T}{\sqrt{13}}$, $T_y = \dfrac{2T}{\sqrt{13}}$.

The weight acts vertically downward at the centre of gravity $G(1.5, 0)$:

$$\vec{W} = -200\,\vec{j}\ \text{N}.$$

The reaction at the hinge A has unknown components:

$$\vec{R} = R_x\,\vec{i} + R_y\,\vec{j}.$$

KEY POINTS **Resolving Forces**

- The component of a vector $\vec{F}$ in the direction of a unit vector $\hat{u}$ is $\vec{F} \cdot \hat{u}$.
- For a force $\vec{F}$ making an angle θ with the horizontal: $F_x = F\cos\theta$, $F_y = F\sin\theta$.
- Always define your coordinate system ($\vec{i}$ horizontal +, $\vec{j}$ vertical +) and stick to it.

2. Moment Equilibrium about Hinge A

We take moments about A (clockwise positive). The forces acting are $\vec{W}$ at G, $\vec{T}$ at B, and $\vec{R}$ at A. The reaction $\vec{R}$ has zero moment about A because its line of action passes through A.

- **Weight $\vec{W}$:** Acts vertically downward at G, horizontal distance $AG = 1.5$ m from A. The perpendicular distance from A to the line of action of $\vec{W}$ is 1.5 m. It tends to rotate the plank clockwise ⇒ positive moment:

$$M_W = +W \times AG = +200 \times 1.5 = +300 \text{ N m}.$$

- **Tension $\vec{T}$:** Only the vertical component $T_y = \frac{2T}{\sqrt{13}}$ produces a moment about A (the horizontal component T_x passes through A). T_y acts upward at B, distance $AB = 3$ m from A, tending to rotate anticlockwise ⇒ negative moment:

$$M_T = -T_y \times AB = -\frac{2T}{\sqrt{13}} \times 3 = -\frac{6T}{\sqrt{13}} \text{ N m}.$$

Equilibrium condition $\sum M_A = 0$:

$$300 - \frac{6T}{\sqrt{13}} = 0 \quad \Rightarrow \quad T = \frac{300\sqrt{13}}{6} = 50\sqrt{13} \text{ N}.$$

Numerically, $\sqrt{13} \approx 3.60555$, so $T \approx 180.3$ N.

WARNING **Moment Calculation**

- The moment of a force about a point is $M = F \times d_\perp$, where $d_\perp$ is the perpendicular distance from the point to the line of action of the force.
- Alternatively, $M = F_\perp \times d$, where $F_\perp$ is the component of the force perpendicular to the line joining the point to the point of application, and d is that distance.
- Choose the sign convention (clockwise + or anticlockwise +) and apply it consistently.

3. Force Equilibrium and Hinge Reaction

Apply $\sum F_x = 0$ and $\sum F_y = 0$.

$\sum F_x = 0$:

$$R_x + T_x = 0 \quad \Rightarrow \quad R_x = -T_x = \frac{3T}{\sqrt{13}} = \frac{3 \times 50\sqrt{13}}{\sqrt{13}} = 150 \text{ N}.$$

$\sum F_y = 0$:

$$R_y + T_y - W = 0 \quad \Rightarrow \quad R_y = W - T_y = 200 - \frac{2T}{\sqrt{13}} = 200 - \frac{2 \times 50\sqrt{13}}{\sqrt{13}} = 200 - 100 = 100 \text{ N}.$$

Hence the reaction vector is

$$\vec{R} = 150\,\vec{i} + 100\,\vec{j} \text{ N}.$$

Its magnitude:

$$R = \sqrt{R_x^2 + R_y^2} = \sqrt{150^2 + 100^2} = \sqrt{32500} = 50\sqrt{13} \approx 180.3 \text{ N}.$$

Its direction (angle α above the horizontal):

$$\tan\alpha = \frac{R_y}{R_x} = \frac{100}{150} = \frac{2}{3} \quad \Rightarrow \quad \alpha = \arctan\left(\frac{2}{3}\right) \approx 33.69^\circ.$$

KEY POINTS **Equilibrium Conditions**

- For a rigid body in static equilibrium in a plane: $\sum F_x = 0, \sum F_y = 0, \sum M_{\text{any point}} = 0$.
- Choosing the moment centre at a point where unknown forces act (like the hinge A) eliminates those unknowns from the moment equation.
- The three forces $\vec{W}$, $\vec{T}$, $\vec{R}$ are not concurrent (their lines of action do not meet at a single point). This is consistent because there are three forces but they are not the only forces? Wait — they are the only forces. For a body in equilibrium under three non-parallel forces, the lines of action must be concurrent. Let's check: $\vec{W}$ is vertical through $G(1.5, 0)$. $\vec{T}$ acts along BC. $\vec{R}$ acts along the line from A at angle α. The lines BC and AR intersect at some point. Does the vertical through G pass through that intersection? In this problem, the three forces are concurrent (a classic result for a simply supported beam with a cable). The intersection of BC and the line of $\vec{R}$ lies on the vertical through G. This can be verified geometrically.

4. Work Done by the Tension for a Small Displacement

The awning is slowly lowered so that B moves vertically downward by $\delta y = 0.01$ m while A remains fixed. The plank rotates about A by a small angle $\delta\phi$. To first order, the displacement of B is purely vertical:

$$\delta\vec{r}_B = -0.01\,\vec{j}\ \text{m}.$$

The work done by a constant force $\vec{T}$ during a displacement $\delta\vec{r}$ is $\delta W = \vec{T}\cdot\delta\vec{r}$.

$$\delta W = \left(-\frac{3T}{\sqrt{13}}\vec{i} + \frac{2T}{\sqrt{13}}\vec{j}\right)\cdot\left(-0.01\,\vec{j}\right) = \frac{2T}{\sqrt{13}} \times (-0.01) = -\frac{0.02T}{\sqrt{13}}\ \text{J}.$$

Substitute $T = 50\sqrt{13}$ N:

$$\delta W = -\frac{0.02 \times 50\sqrt{13}}{\sqrt{13}} = -1.0\ \text{J}.$$

KEY POINTS **Work Done by a Constant Force**

- $W = \vec{F}\cdot\vec{d} = Fd\cos\phi$, where ϕ is the angle between $\vec{F}$ and $\vec{d}$.
- In component form: $W = F_x d_x + F_y d_y$.
- Negative work means the force component opposes the displacement; energy is transferred from the body to the force source (or stored as potential energy in the ca-

ble). Here, the tension pulls upward while B moves downward, so the tension does negative work. An external agent must do positive work to lower the awning.

5. Adding a Second Symmetric Cable: Couple and Concurrency Analysis

A second identical cable is attached from B to $D(0, -2)$ (vertically 2 m below A). The two cables make equal angles $\theta = \arctan(2/3)$ with the horizontal, one above, one below. By symmetry, the tensions have equal magnitude T' (to be determined). Their vectors are:

$$\vec{T}_1 = T'\left(-\frac{3}{\sqrt{13}}\vec{i} + \frac{2}{\sqrt{13}}\vec{j}\right), \qquad \vec{T}_2 = T'\left(-\frac{3}{\sqrt{13}}\vec{i} - \frac{2}{\sqrt{13}}\vec{j}\right).$$

1. **Do $\vec{T}_1$ and $\vec{T}_2$ form a couple?** A **couple** consists of two forces that are:
   - equal in magnitude,
   - opposite in direction (parallel lines of action, anti-parallel vectors),
   - not collinear (different lines of action).

   Here, $\vec{T}_1$ and $\vec{T}_2$ have the same horizontal component (both leftward) and opposite vertical components. They are **not** opposite vectors; their resultant is a pure horizontal force:

   $$\vec{T}_1 + \vec{T}_2 = -2T'\frac{3}{\sqrt{13}}\vec{i} = -\frac{6T'}{\sqrt{13}}\vec{i}.$$

   Since the resultant force is not zero, they do **not** form a couple. They are equivalent to a single horizontal force acting at B.

2. **Three-force concurrency condition.** The forces on the plank are now $\vec{W}$ (at G), $\vec{T}_1$ (along BC), $\vec{T}_2$ (along BD), and $\vec{R}$ (at A) — four forces. The three-force concurrency theorem applies only when exactly three forces act. If we consider the resultant of the two cables, $\vec{T}_{\text{res}} = \vec{T}_1 + \vec{T}_2$ (horizontal at B), then we have three forces: $\vec{W}$, $\vec{T}_{\text{res}}$, $\vec{R}$. For equilibrium, their lines of action must be concurrent. $\vec{T}_{\text{res}}$ is horizontal through B; $\vec{W}$ is vertical through G; they intersect at $(3, 0)$? No, $\vec{W}$ is at $x = 1.5$, $\vec{T}_{\text{res}}$ is at $y = 0$. They intersect at $(1.5, 0)$? Wait: $\vec{T}_{\text{res}}$ acts at $B(3, 0)$ horizontally. Its line of action is the x-axis ($y = 0$). $\vec{W}$ acts at $G(1.5, 0)$ vertically. Its line of action is $x = 1.5$. These two lines intersect at $(1.5, 0) = G$. The reaction $\vec{R}$ at $A(0, 0)$ must also pass through this intersection for concurrency. But $\vec{R}$ passes through $A(0, 0)$ and $G(1.5, 0)$ only if it is horizontal. However, $\vec{R}$ must balance the vertical weight, so it has a vertical component. Thus the three lines are **not concurrent** — confirming that equilibrium is impossible with only these three forces (or four forces with a hinge that provides no moment). The hinge reaction $\vec{R}$ provides the necessary fourth force to satisfy equilibrium, but as we see next, even four forces cannot satisfy moment equilibrium with this geometry.

3. **Moment equilibrium with two cables (viability check).** Take moments about A again. The forces with moments about A are $\vec{W}$ and the vertical components of $\vec{T}_1$ and $\vec{T}_2$ (horizontal

components pass through A).

$$M_W = +W \times 1.5 = +300\ \text{N m} \quad \text{(clockwise)}$$
$$M_{T_1} = -T'_y \times 3 = -\frac{2T'}{\sqrt{13}} \times 3 = -\frac{6T'}{\sqrt{13}}\ \text{N m} \quad \text{(anticlockwise)}$$
$$M_{T_2} = +T'_y \times 3 = +\frac{2T'}{\sqrt{13}} \times 3 = +\frac{6T'}{\sqrt{13}}\ \text{N m} \quad \text{(clockwise, since } T_{2y} \text{ is downward)}$$

The moments of the two vertical components **cancel exactly** ($M_{T_1} + M_{T_2} = 0$). The weight moment +300 N m remains unbalanced. The hinge A cannot provide a moment (it is a simple hinge). Therefore:

$$\sum M_A = 300 \neq 0.$$

Conclusion: The symmetric two-cable configuration **cannot be in static equilibrium** with a simple hinge at A. The awning would rotate clockwise. To make it work, one would need:

- A fixed support (built-in end) at A that can exert a reaction moment, or
- The second cable attached at a different point on the plank (not at B), or
- The plank not horizontal (inclined).

DID YOU KNOW? **Real-World Design Insight**

In practice, market stall awnings in Cameroon often use a single inclined strut (a compression member) or a single cable with the hinge providing both horizontal and vertical reaction. Adding a second symmetric cable at the same point creates a "redundant" constraint that fights the weight's moment instead of helping it. Engineers use static determinacy checks: a planar rigid body has 3 equilibrium equations. A hinge (2 reactions) + 1 cable (1 tension) = 3 unknowns → statically determinate (solvable). Hinge + 2 cables = 4 unknowns → statically indeterminate; but here the geometry makes it impossible to satisfy all equations simultaneously — a mechanism or improper constraint.

7.4 Safety Assessment and Conclusion

Component	Calculated Value	Maximum Allowable	Status
Cable Tension T	$50\sqrt{13} \approx 180.3$ N	500 N	SAFE (Margin ≈ 2.8×)
Hinge Reaction R	$50\sqrt{13} \approx 180.3$ N	400 N	SAFE (Margin ≈ 2.2×)

The single-cable design meets the safety requirements comfortably. The proposed symmetric two-cable modification is **not viable** with a simple hinge because it cannot satisfy moment equilibrium. A different arrangement (e.g., attaching the second cable to a point on the plank between A and B, or using a fixed support at A) would be needed for a two-cable system.

7.5 Key Points for Examination Revision

KEY POINTS **Essential Summary**

- Free-body diagram is the starting point: identify all forces, their points of application, and directions.
- Vector components simplify equilibrium: $\sum F_x = 0, \sum F_y = 0, \sum M = 0$.
- Moment equilibrium about a convenient point (often a hinge) eliminates unknown reactions.
- Work $W = \vec{F} \cdot \vec{d}$; negative work means the force opposes the displacement.
- Couple: two equal, opposite, parallel forces with different lines of action → pure moment, no resultant force.
- Three-force concurrency: if a rigid body is in equilibrium under three non-parallel forces, their lines of action must meet at a single point.
- Statically determinate systems have exactly as many unknowns as independent equilibrium equations.
- Always check safety margins against material limits.

7.6 Common Pitfalls to Avoid

WARNING **Examiner's Traps**

- Forgetting that the weight of a uniform plank acts at its centre, not at the end.
- Using the wrong perpendicular distance when calculating moments (always the perpendicular distance from the pivot to the line of action of the force).
- Confusing the sign convention for moments (clockwise vs anticlockwise) and for work (positive when force and displacement are in the same general direction).
- Assuming two symmetric forces automatically form a couple; they must be equal and opposite (anti-parallel).
- Neglecting that a hinge can exert a reaction force but no moment.
- In the two-cable scenario, assuming the tension magnitude remains the same as in the single-cable case. The equilibrium equations change, so the tensions change (or equilibrium becomes impossible).

EXERCISE 6 **Practice Problem**

A uniform rod AB of length 4 m and weight 150 N is hinged at A to a vertical wall. It is held horizontal by a light inextensible string attached at B and to a point C on the wall 3 m above A. A load of 100 N is suspended from the midpoint of the rod. Find:

1. The tension in the string.
2. The magnitude and direction of the reaction at the hinge.
3. The work done by the tension if the rod is lowered until B descends 0.02 m.

SOLUTION **Practice Problem**

1. Geometry: $AB = 4$, $AC = 3$, so $BC = 5$, $\sin\theta = 3/5$, $\cos\theta = 4/5$. Moments about A: $150 \times 2 + 100 \times 2 = T\sin\theta \times 4 \Rightarrow 500 = T \times \frac{3}{5} \times 4 = \frac{12T}{5} \Rightarrow T = \frac{2500}{12} = \frac{625}{3} \approx 208.3\,\text{N}$.
2. $R_x = T\cos\theta = \frac{625}{3} \times \frac{4}{5} = \frac{500}{3} \approx 166.7\,\text{N}$. $R_y = 150 + 100 - T\sin\theta = 250 - \frac{625}{3} \times \frac{3}{5} = 250 - 125 = 125\,\text{N}$. $R = \sqrt{(500/3)^2 + 125^2} = \sqrt{250000/9 + 15625} = \sqrt{390625/9} = 625/3 \approx 208.3\,\text{N}$. $\tan\alpha = 125/(500/3) = 3/4 \Rightarrow \alpha \approx 36.87^\circ$.
3. $\delta\vec{r}_B = -0.02\,\vec{j}$. $\vec{T} = T(-\cos\theta\,\vec{i} + \sin\theta\,\vec{j}) = \frac{625}{3}(-\frac{4}{5}\vec{i} + \frac{3}{5}\vec{j})$. $\delta W = \vec{T}\cdot\delta\vec{r}_B = \frac{625}{3} \times \frac{3}{5} \times (-0.02) = 125 \times (-0.02) = -2.5\,\text{J}$.

8 METHODS AND WORKED EXAMPLES

9 APPLICATION OF SCALAR AND VECTOR PRODUCTS

9.1 Vector Component of a Vector in a Given Direction

DEFINITION **Scalar and Vector Components**

Let $\vec{a}$ and $\vec{b}$ be two vectors, with $\vec{b} \neq \vec{0}$. The **scalar component** (or **projection**) of $\vec{a}$ onto the direction of $\vec{b}$ is the signed magnitude

$$a_b = \vec{a} \cdot \hat{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|},$$

where $\hat{b} = \dfrac{\vec{b}}{|\vec{b}|}$ is the unit vector in the direction of $\vec{b}$. The **vector component** of $\vec{a}$ parallel to $\vec{b}$ is

$$\vec{a}_{\parallel} = a_b \hat{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}.$$

The perpendicular component is $\vec{a}_{\perp} = \vec{a} - \vec{a}_{\parallel}$; it satisfies $\vec{a}_{\parallel} \cdot \vec{a}_{\perp} = 0$ and $\vec{a}_{\parallel} + \vec{a}_{\perp} = \vec{a}$.

PROPERTY **Geometric meaning**

If θ is the angle between $\vec{a}$ and $\vec{b}$, then $a_b = |\vec{a}| \cos\theta$. The sign of a_b tells whether the projection points along $\vec{b}$ ($a_b > 0$) or opposite to $\vec{b}$ ($a_b < 0$). The magnitude of the perpendicular component is $|\vec{a}_{\perp}| = |\vec{a}| \sin\theta$.

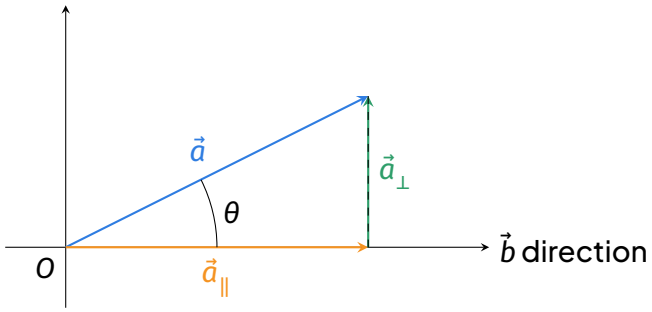
METHOD **Component of a vector along a given direction**

To find the vector component of $\vec{a}$ in the direction of $\vec{b}$:

1. Compute the unit vector $\hat{b} = \dfrac{\vec{b}}{|\vec{b}|}$.
2. Compute the scalar component (projection): $a_b = \vec{a} \cdot \hat{b} = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.

3. Compute the vector component: $\vec{a}_{\parallel} = (\vec{a} \cdot \hat{b})\,\hat{b} = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\,\vec{b}$.
4. If the perpendicular component is needed: $\vec{a}_{\perp} = \vec{a} - \vec{a}_{\parallel}$.
5. Check: $\vec{a}_{\parallel} \cdot \vec{a}_{\perp} = 0$ and $\vec{a}_{\parallel} + \vec{a}_{\perp} = \vec{a}$.

Reflex: if the scalar component is negative, the projection points opposite to $\vec{b}$.



Decomposition of $\vec{a}$ into components parallel and perpendicular to $\vec{b}$.

WORKED EXAMPLE 15 **Resolving a displacement along a road**

A surveyor in Bamenda moves from point A to point B, with displacement $\vec{s} = \begin{pmatrix} 6 \\ 2 \\ -3 \end{pmatrix}$ km. A straight road runs in the direction $\vec{d} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$. Find (a) the scalar component of $\vec{s}$ along the road; (b) the vector component of $\vec{s}$ along the road; (c) the perpendicular distance from B to the road direction through A.

SOLUTION

(a) Scalar component. First $|\vec{d}| = \sqrt{4 + 1 + 4} = 3$, so $\hat{d} = \dfrac{1}{3}\begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

$$s_d = \vec{s} \cdot \hat{d} = \frac{1}{3}(12 - 2 - 6) = \frac{4}{3}\ \text{km}.$$

(b) Vector component.

$$\vec{s}_{\parallel} = \frac{4}{3}\hat{d} = \frac{4}{9}\begin{pmatrix}2\\-1\\2\end{pmatrix} = \begin{pmatrix}8/9\\-4/9\\8/9\end{pmatrix}\text{ km}.$$

(c) Perpendicular component.

$$\vec{s}_{\perp} = \vec{s} - \vec{s}_{\parallel} = \begin{pmatrix}6 - 8/9\\2 + 4/9\\-3 - 8/9\end{pmatrix} = \begin{pmatrix}46/9\\22/9\\-35/9\end{pmatrix}.$$

Its magnitude:

$$|\vec{s}_{\perp}| = \frac{1}{9}\sqrt{46^2 + 22^2 + 35^2} = \frac{1}{9}\sqrt{2116 + 484 + 1225} = \frac{\sqrt{3825}}{9} = \frac{15\sqrt{17}}{9} = \frac{5\sqrt{17}}{3} \approx 6.87\text{ km}.$$

Check: $\vec{s}_{\parallel} \cdot \vec{s}_{\perp} = \frac{1}{81}(8 \times 46 - 4 \times 22 - 8 \times 35) = \frac{368 - 88 - 280}{81} = 0$. ✓

9.2 Work Done by a Constant Force

DEFINITION **Work Done by a Constant Force**

If a constant force $\vec{F}$ acts on a particle that undergoes a displacement $\vec{s}$, the **work done** by the force is the scalar product

$$W = \vec{F} \cdot \vec{s} = |\vec{F}|\,|\vec{s}|\cos\theta,$$

where θ is the angle between $\vec{F}$ and $\vec{s}$. The SI unit is the joule (J): $1\,\text{J} = 1\,\text{N} \cdot \text{m}$.

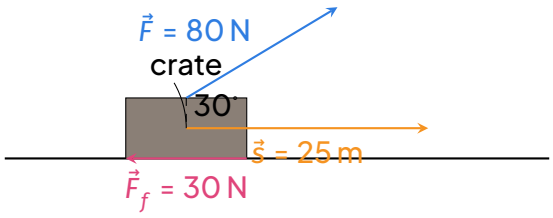
PROPERTY **Sign of work**

- $W > 0$: the force has a component in the direction of displacement (force helps the motion).
- $W = 0$: the force is perpendicular to the displacement (e.g. weight on horizontal ground, normal reaction).
- $W < 0$: the force has a component opposite to the displacement (force opposes the motion, e.g. friction).

METHOD **Computing work done**

1. Identify the force $\vec{F}$ (in newtons) and the displacement $\vec{s}$ (in metres) as vectors.
2. Work done: $W = \vec{F} \cdot \vec{s} = |\vec{F}||\vec{s}| \cos\theta$ (in joules).
3. If only magnitudes and the angle θ between force and displacement are known, use the trigonometric form directly.
4. Interpret the sign: $W > 0$ (force helps motion), $W = 0$ (force perpendicular to motion), $W < 0$ (force opposes motion, e.g. friction).
5. For several forces, add the works algebraically, or sum the forces first: $W = \left(\sum \vec{F}_i\right) \cdot \vec{s}$.

Reflex: weight does no work on horizontal ground; friction always does negative work.



Force inclined at 30° pulling a crate; friction opposes motion.

WORKED EXAMPLE 16 **Work done pulling a load**

A porter at the Douala market pulls a crate along horizontal ground with a rope inclined at 30° to the horizontal, exerting a constant force of 80 N. The crate moves 25 m. (a) Find the work done by the porter. (b) A friction force of 30 N opposes the motion; find the total work done on the crate. (c) Using unit vectors $\vec{i}$ (horizontal) and $\vec{j}$ (vertical), write $\vec{F}$ and $\vec{s}$ and verify part (a) with the scalar product.

SOLUTION

(a) $W = |\vec{F}||\vec{s}| \cos\theta = 80 \times 25 \times \cos 30^\circ = 2000 \times \dfrac{\sqrt{3}}{2} = 1000\sqrt{3} \approx 1732$ J. (b) Work done by friction: $W_f = 30 \times 25 \times \cos 180^\circ = -750$ J. Weight and normal reaction do no work (perpendicular to motion). Total work: $W_{\text{tot}} = 1000\sqrt{3} - 750 \approx 982$ J. (c) $\vec{F} = 80\cos 30^\circ\,\vec{i} + 80\sin 30^\circ\,\vec{j} = 40\sqrt{3}\,\vec{i} + 40\,\vec{j}$ N; $\vec{s} = 25\,\vec{i}$ m.

$$W = \vec{F} \cdot \vec{s} = 40\sqrt{3} \times 25 + 40 \times 0 = 1000\sqrt{3}\text{ J}.$$

WORKED EXAMPLE 17 **Work done in three dimensions**

A constant force $\vec{F} = (2\vec{i} - 3\vec{j} + 6\vec{k})$ N moves a particle from $P(1, 0, 2)$ to $Q(4, 5, -1)$, coordinates in metres. Find the work done.

SOLUTION

Displacement: $\vec{s} = \vec{PQ} = \begin{pmatrix} 4-1 \\ 5-0 \\ -1-2 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ -3 \end{pmatrix}$ m.

$$W = \vec{F} \cdot \vec{s} = (2)(3) + (-3)(5) + (6)(-3) = 6 - 15 - 18 = -27 \text{ J}.$$

The negative sign means the force globally opposes the displacement: 27 J of work must be supplied against the force.

9.3 Moment of a Force and Equilibrium

DEFINITION **Moment of a Force about a Point**

The **moment** of a force $\vec{F}$ about a point O is the vector

$$\vec{M}_O = \vec{r} \times \vec{F},$$

where $\vec{r}$ is the position vector from O to any point on the line of action of $\vec{F}$. Its magnitude is $M = Fd$, where d is the perpendicular distance from O to the line of action. In two dimensions, we use the scalar moment $M = Fd$ with a sign convention (usually anticlockwise positive).

PROPERTY **Equilibrium of a Rigid Body**

A rigid body is in equilibrium under a system of forces if and only if

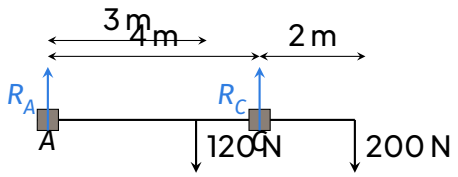
$$\sum \vec{F} = \vec{0} \quad \text{and} \quad \sum \vec{M}_O = \vec{0} \text{ about any point } O.$$

In two dimensions this gives three independent scalar equations: $\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$.

METHOD **Moment about a point; equilibrium**

1. The moment of $\vec{F}$ about point O is $\vec{M}_O = \vec{r} \times \vec{F}$, where $\vec{r}$ is the position vector from O to any point of the line of action.
2. In 2D, use the scalar moment: $M = Fd$, where d is the **perpendicular distance** from O to the line of action; fix a positive sense (usually anticlockwise).
3. For **equilibrium** of a rigid body: $\sum \vec{F} = \vec{0}$ and $\sum \vec{M}_O = \vec{0}$ about any point O.
4. Strategy: take moments about a point where unknown forces act, to eliminate them.
5. In 3D, compute $\vec{r} \times \vec{F}$ with the determinant formula and set each component of the total moment to zero.

Reflex: a force whose line of action passes through O has zero moment about O.



Beam AB with supports at A and C, weight at midpoint, load at B.

WORKED EXAMPLE 18 **Equilibrium of a uniform beam**

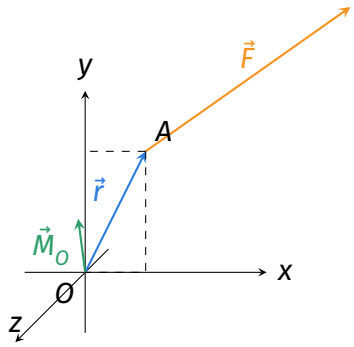
A uniform beam AB of length 6 m and weight 120 N rests horizontally on two supports at A and at C, where $AC = 4$ m. A load of 200 N hangs at B. Find the reactions at the supports.

SOLUTION

Let R_A and R_C be the vertical reactions. The weight acts at the midpoint, 3 m from A. Vertical equilibrium: $R_A + R_C = 120 + 200 = 320$ N. Moments about A (anticlockwise positive):

$$R_C \times 4 - 120 \times 3 - 200 \times 6 = 0 \implies 4R_C = 360 + 1200 = 1560 \implies R_C = 390 \text{ N}.$$

Then $R_A = 320 - 390 = -70$ N. Interpretation: $R_A < 0$ means the support at A would have to pull down: the beam tips about C unless it is bolted at A. The load is too far beyond C for a simple resting contact at A.



Moment $\vec{M}_O$ = $\vec{r}$ × $\vec{F}$ about the origin.

WORKED EXAMPLE 19 **Moment as a vector product**

A force $\vec{F} = (3\vec{i} + 2\vec{j} - \vec{k})$ N acts at the point $A(1, 2, 0)$, coordinates in metres. Find the moment of $\vec{F}$ about the origin O, and its magnitude.

SOLUTION

$\vec{r} = \vec{OA} = \vec{i} + 2\vec{j}$.

$$\vec{M}_O = \vec{r} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 0 \\ 3 & 2 & -1 \end{vmatrix} = \vec{i}(2 \times (-1) - 0 \times 2) - \vec{j}(1 \times (-1) - 0 \times 3) + \vec{k}(1 \times 2 - 2 \times 3)$$

$$= -2\vec{i} + \vec{j} - 4\vec{k} \text{ N m}.$$

Magnitude: $|\vec{M}_O| = \sqrt{4 + 1 + 16} = \sqrt{21} \approx 4.58$ N m.

9.4 Single Forces and Couples

DEFINITION **Couple**

A **couple** consists of two forces equal in magnitude, opposite in direction, parallel but not collinear. The resultant force is zero, but the moment (torque) is non-zero and independent of the reference point.

PROPERTY **Moment of a couple**

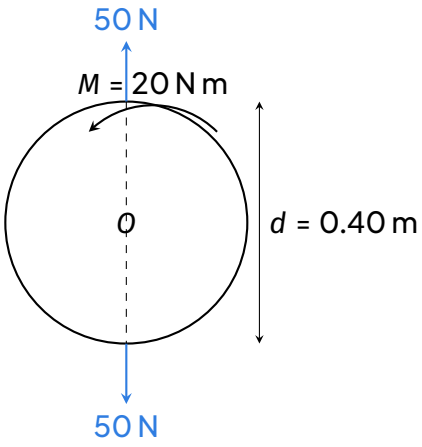
The moment of a couple is $M = Fd$, where F is the magnitude of one force and d is the per-

pendicular distance between their lines of action. The moment vector $\vec{G}$ is the same about every point. A couple cannot be replaced by a single force.

METHOD **Resultant force and couples**

1. A **couple** is a pair of equal, opposite, parallel forces not in the same line: resultant force zero, but a non-zero moment.
2. Moment of a couple: $M = Fd$, where d is the perpendicular distance between the two lines of action. It is the same about every point.
3. To reduce a system to a single force at a point O: resultant $\vec{R} = \sum \vec{F}_i$ plus a couple of moment $\vec{G}_O = \sum \vec{M}_O(\vec{F}_i)$.
4. A single force $\vec{F}$ acting at A is equivalent to the same force at O plus a couple of moment $\vec{OA} \times \vec{F}$.
5. Two systems are equivalent if and only if they have the same resultant force and the same total moment about one (hence any) point.

Reflex: a couple can never be replaced by a single force.



Couple on a steering wheel: equal opposite forces, moment Fd.

WORKED EXAMPLE 20 **Identifying a couple**

Two horizontal forces of 50 N act on a steering wheel of diameter 40 cm: one pushes north at the top of the wheel, the other pushes south at the bottom. (a) Show the resultant force

is zero. (b) Find the moment of the couple. (c) Verify the moment is the same about the centre O and about the top point T.

SOLUTION

(a) The forces are equal (50 N), parallel and opposite: $\vec{R} = 50 - 50 = 0$ N. (b) $M = Fd = 50 \times 0.40 = 20$ N m. (c) About O: each force has lever arm 0.20 m and both turn the wheel the same way: $M_O = 50(0.20) + 50(0.20) = 20$ N m. About T: the force at T has zero moment; the force at the bottom has lever arm 0.40 m: $M_T = 50 \times 0.40 = 20$ N m. Same value, as expected for a couple. ✓

WORKED EXAMPLE 21 **Replacing a force by a force and a couple**

A force $\vec{F} = (4\vec{i} - 3\vec{j})$ N acts at $A(2, 1)$. Replace it by an equal force through the origin O together with a couple.

SOLUTION

The force through O is $\vec{F} = 4\vec{i} - 3\vec{j}$ N. The couple has moment

$$\vec{G} = \vec{OA} \times \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 1 & 0 \\ 4 & -3 & 0 \end{vmatrix} = \vec{k}(2 \times (-3) - 1 \times 4) = -10\,\vec{k} \text{ N m}.$$

So the system is: force $(4\vec{i} - 3\vec{j})$ N at O, plus a clockwise couple of moment 10 N m.

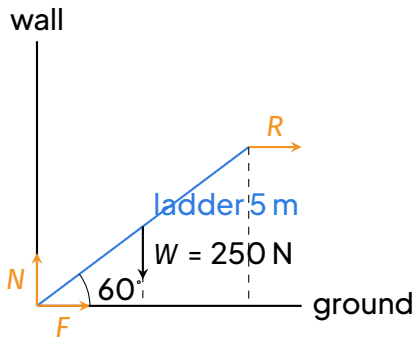
9.5 Coplanar Forces and Equilibrium

METHOD **Solving coplanar equilibrium problems**

1. Draw a clear force diagram: mark all forces (weights, reactions, tensions, friction).
2. Choose perpendicular axes (often horizontal/vertical or along/perpendicular to a plane) and resolve: $\sum F_x = 0, \sum F_y = 0$.
3. Take moments about a well-chosen point: $\sum M = 0$. Three independent equations for a rigid body in a plane.
4. For three forces only in equilibrium: they are concurrent or parallel — use the **triangle of forces** or Lami's theorem: $\dfrac{F_1}{\sin \alpha_1} = \dfrac{F_2}{\sin \alpha_2} = \dfrac{F_3}{\sin \alpha_3}$, where α_i is the angle between the other two forces.

5. Solve the simultaneous equations and check signs physically.

Reflex: with exactly three forces, Lami's theorem is usually the fastest route.



Ladder against smooth wall, rough ground.

WORKED EXAMPLE 22 **Ladder against a wall**

A uniform ladder of weight $W = 250$ N and length 5 m leans against a smooth vertical wall, its foot on rough horizontal ground, making 60° with the ground. Find the reaction of the wall, and the vertical and frictional components of the ground reaction.

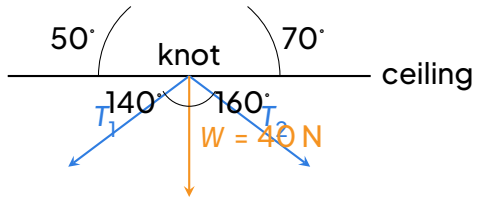
SOLUTION

Let R = wall reaction (horizontal, wall smooth), N = normal ground reaction, F = friction at the ground. The weight acts at the midpoint. Resolve horizontally: $F - R = 0 \implies F = R$. Resolve vertically: $N - 250 = 0 \implies N = 250$ N. Moments about the foot (lever arms: weight → $2.5\cos 60^\circ$; wall reaction → $5\sin 60^\circ$):

$$R \times 5\sin 60^\circ - 250 \times 2.5\cos 60^\circ = 0$$

$$R = \frac{250 \times 2.5 \times 0.5}{5 \times \frac{\sqrt{3}}{2}} = \frac{312.5}{4.330} = \frac{125}{2\sqrt{3}} = \frac{125\sqrt{3}}{6} \approx 36.1\,\text{N}.$$

Hence $F \approx 36.1$ N, $N = 250$ N, $R \approx 36.1$ N.



Three forces in equilibrium at the knot: Lami's theorem applies.

WORKED EXAMPLE 23 **Three forces: Lami's theorem**

A picture of weight 40 N hangs by two strings from two hooks. The strings make angles 50° and 70° with the horizontal ceiling. Find the tension in each string.

SOLUTION

Three forces in equilibrium: T_1, T_2, $W = 40$ N. The angles between the forces at the knot: between T_1 and T_2: $180^\circ - 50^\circ - 70^\circ = 60^\circ$; between T_1 and the vertical weight: $90^\circ + 50^\circ = 140^\circ$; between T_2 and the weight: $90^\circ + 70^\circ = 160^\circ$. Lami's theorem:

$$\frac{T_1}{\sin 160^\circ} = \frac{T_2}{\sin 140^\circ} = \frac{40}{\sin 60^\circ}.$$

$$T_1 = \frac{40 \sin 160^\circ}{\sin 60^\circ} = \frac{40 \times 0.3420}{0.8660} \approx 15.8 \text{ N}, \qquad T_2 = \frac{40 \sin 140^\circ}{\sin 60^\circ} = \frac{40 \times 0.6428}{0.8660} \approx 29.7 \text{ N}.$$

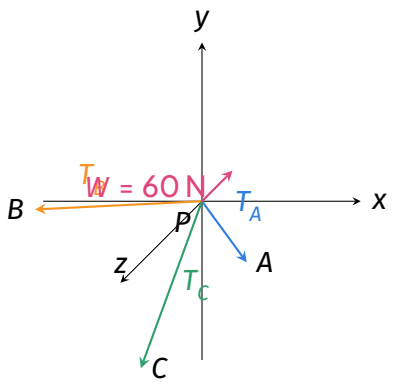
Check (horizontal): $T_2 \cos 70^\circ \approx 29.7 \times 0.3420 \approx 10.2$ N and $T_1 \cos 50^\circ \approx 15.8 \times 0.6428 \approx 10.2$ N. ✓

9.6 Simple Systems of Forces in Three Dimensions

METHOD **Equilibrium in 3D**

1. Express every force in component form $\vec{F}_i = F_{ix}\vec{i} + F_{iy}\vec{j} + F_{iz}\vec{k}$; for a force along a line AB, use $\vec{F} = F\,\dfrac{\vec{AB}}{|\vec{AB}|}$.
2. Resultant: $\vec{R} = \sum \vec{F}_i$. Equilibrium of a particle: $\vec{R} = \vec{0}$, i.e. three scalar equations.
3. For a rigid body, add the moment condition: $\sum \vec{r}_i \times \vec{F}_i = \vec{0}$ (three more scalar equations).
4. Solve the system of up to six scalar equations for the unknowns.

Reflex: write each force as magnitude × unit vector along its line — this is where most marks are won or lost.



Three cables supporting a lamp at $P(0,0,0)$; anchors at $z = 3$ m.

WORKED EXAMPLE 24 **Particle held by three cables**

A lamp of weight 60 N hangs at point $P(0,0,0)$ supported by three cables anchored at $A(2,0,3)$, $B(-2,1,3)$ and $C(0,-2,3)$ (units: metres). Find the tension in each cable, given symmetry is broken and the tensions are T_A, T_B, T_C.

SOLUTION

Unit vectors from P to each anchor:

$$\vec{PA} = \begin{pmatrix}2\\0\\3\end{pmatrix},\ |\vec{PA}| = \sqrt{13};\quad \vec{PB} = \begin{pmatrix}-2\\1\\3\end{pmatrix},\ |\vec{PB}| = \sqrt{14};\quad \vec{PC} = \begin{pmatrix}0\\-2\\3\end{pmatrix},\ |\vec{PC}| = \sqrt{13}.$$

Equilibrium $\sum \vec{F} = \vec{0}$ gives:

$$\begin{aligned} x:&\quad \frac{2T_A}{\sqrt{13}} - \frac{2T_B}{\sqrt{14}} = 0,\\ y:&\quad \frac{T_B}{\sqrt{14}} - \frac{2T_C}{\sqrt{13}} = 0,\\ z:&\quad \frac{3T_A}{\sqrt{13}} + \frac{3T_B}{\sqrt{14}} + \frac{3T_C}{\sqrt{13}} = 60. \end{aligned}$$

From the x-equation: $T_B = \dfrac{\sqrt{14}}{\sqrt{13}}T_A$. From the y-equation: $T_C = \dfrac{\sqrt{13}}{2\sqrt{14}}T_B = \dfrac{T_A}{2}$. Substitute into the z-equation:

$$\frac{3T_A}{\sqrt{13}} + \frac{3T_A}{\sqrt{13}} + \frac{3T_A}{2\sqrt{13}} = 60 \implies \frac{3T_A}{\sqrt{13}}\left(1+1+\tfrac{1}{2}\right) = 60 \implies \frac{15T_A}{2\sqrt{13}} = 60.$$

$$T_A = \frac{120\sqrt{13}}{15} = 8\sqrt{13} \approx 28.8\ \text{N},\quad T_C = 4\sqrt{13} \approx 14.4\ \text{N},\quad T_B = 8\sqrt{14} \approx 29.9\ \text{N}.$$

Check (z): $\frac{3}{\sqrt{13}}(8\sqrt{13}+4\sqrt{13})+\frac{3}{\sqrt{14}}(8\sqrt{14}) = 36+24 = 60$ N. ✓

KEY POINTS **Key formulas for the chapter**

- Scalar component of $\vec{a}$ on $\vec{b}$: $a_b = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|}$.
- Vector component: $\vec{a}_{\parallel} = \frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\vec{b}$; perpendicular: $\vec{a}_{\perp} = \vec{a} - \vec{a}_{\parallel}$.
- Work: $W = \vec{F}\cdot\vec{s} = Fs\cos\theta$ (joules).
- Moment about O: $\vec{M}_O = \vec{r}\times\vec{F}$; magnitude $M = Fd$ (d = perpendicular distance).
- Equilibrium: $\sum\vec{F} = \vec{0}, \sum\vec{M}_O = \vec{0}$.
- Couple moment: $M = Fd$ (same about any point).
- Lami's theorem (3 forces): $\frac{F_1}{\sin\alpha_1} = \frac{F_2}{\sin\alpha_2} = \frac{F_3}{\sin\alpha_3}$.
- 3D force along AB: $\vec{F} = F\frac{\vec{AB}}{|\vec{AB}|}$.

DID YOU KNOW? **From Cameroon to the stars**

The principles of moments and equilibrium are used daily in Cameroonian engineering: from designing the supports of the Pont de la Réunification in Douala to calculating the tension in the guy-wires of telecommunication masts on Mount Cameroon. In space, the same vector methods control the orientation of satellites — reaction wheels produce couples to turn the spacecraft without using fuel. The scalar product giving work is the foundation of the work–energy theorem, which explains why a car climbing the steep roads of Buea needs more fuel than on the flat Littoral plains.

WARNING **Frequent errors at the GCE**

- Confusing the scalar component (a number) with the vector component (a vector).
- Using $\sin\theta$ instead of $\cos\theta$ in $W = Fs\cos\theta$ — θ is the angle between force and displacement.

- Taking moments with the distance along the beam instead of the perpendicular distance to the line of action.
- Forgetting that the moment of a couple is independent of the point chosen.
- In 3D, forgetting to divide by the length when forming unit vectors along cables.
- Misidentifying the angles in Lami's theorem: α_i is the angle between the other two forces, not the angle the force makes with the horizontal.
- Writing $\vec{r} \times \vec{F}$ with $\vec{r}$ from the force to the point instead of from the point to the force.

EXERCISE 7 **Practice problems**

1. Find the vector component of $\vec{a} = 3\vec{i} - 4\vec{j} + 12\vec{k}$ parallel to $\vec{b} = 2\vec{i} + \vec{j} - 2\vec{k}$, and the perpendicular distance from the tip of $\vec{a}$ to the line of $\vec{b}$.

2. A force $\vec{F} = 5\vec{i} + 12\vec{j}$ N moves an object from $(1, 2)$ to $(4, 6)$ (metres). Find the work done. If a friction force $\vec{F}_f = -3\vec{i} - 4\vec{j}$ N acts, find the net work.

3. A uniform rod AB of weight W and length L rests with A on rough horizontal ground and B against a smooth vertical wall. The rod makes angle θ with the ground. Show that the minimum coefficient of friction for equilibrium is $\mu_{\min} = \tfrac{1}{2}\cot\theta$.

4. Two forces $\vec{F}_1 = 10\vec{j}$ N and $\vec{F}_2 = -10\vec{j}$ N act at $(0, 0, 0)$ and $(0, 0, 0.5)$ respectively. Find the moment of this couple about the origin and about the point $(1, 2, 3)$.

5. A sign of weight 200 N hangs from a horizontal beam AB of length 2 m, hinged at A to a vertical wall. A cable from B to a point C on the wall 1.5 m above A holds the beam horizontal. Find the tension in the cable and the reaction at the hinge.

6. Four forces act on a particle at the origin: $\vec{F}_1 = 20\vec{i}$ N, $\vec{F}_2 = 15\vec{j}$ N, $\vec{F}_3$ along the line to $(3, 4, 0)$ with magnitude 25 N, and $\vec{F}_4$ along the line to $(-1, 2, 2)$ with magnitude 3 N. Is the particle in equilibrium? If not, find the additional force needed.

SOLUTION **Exercise 1**

$\vec{a}\cdot\vec{b} = 6 - 4 - 24 = -22$, $|\vec{b}| = 3$, $\hat{b} = \tfrac{1}{3}(2, 1, -2)$. Scalar component $a_b = -22/3$. Vector component $\vec{a}_{\parallel} = \tfrac{-22}{9}(2, 1, -2) = \left(-\tfrac{44}{9}, -\tfrac{22}{9}, \tfrac{44}{9}\right)$. $\vec{a}_{\perp} = \vec{a} - \vec{a}_{\parallel} = \left(\tfrac{71}{9}, -\tfrac{14}{9}, \tfrac{64}{9}\right)$. Distance $|\vec{a}_{\perp}| = \tfrac{1}{9}\sqrt{71^2 + 14^2 + 64^2} = \tfrac{\sqrt{9213}}{9} \approx 10.66$.

SOLUTION **Exercise 2**

$\vec{s} = (3, 4)$, $W = \vec{F} \cdot \vec{s} = 15 + 48 = 63$ J. $\vec{F}_{\text{net}} = (2, 8)$, $W_{\text{net}} = 6 + 32 = 38$ J. Alternatively $W_f = \vec{F}_f \cdot \vec{s} = -9 - 16 = -25$ J, net $= 63 - 25 = 38$ J.

SOLUTION **Exercise 3**

Forces: N_A (up), F_A (right), N_B (left), W (down at centre). Horizontal: $F_A = N_B$. Vertical: $N_A = W$. Moments about A: $N_B L \sin\theta - W\frac{L}{2}\cos\theta = 0 \Rightarrow N_B = \frac{W}{2}\cot\theta$. Then $F_A = \frac{W}{2}\cot\theta$. Friction limit: $F_A \le \mu N_A \Rightarrow \frac{W}{2}\cot\theta \le \mu W \Rightarrow \mu \ge \frac{1}{2}\cot\theta$.

SOLUTION **Exercise 4**

Couple moment $\vec{G} = \vec{r} \times \vec{F}$ where $\vec{r}$ is from any point on $\vec{F}_1$ to any point on $\vec{F}_2$. Take $\vec{r} = (0, 0, 0.5)$. $\vec{G} = (0, 0, 0.5) \times (0, 10, 0) = (-5, 0, 0)$ N m. Magnitude 5 N m. About $(1, 2, 3)$: $\vec{r}_1 = (-1, -2, -3)$, $\vec{r}_2 = (-1, -2, -2.5)$. $\vec{M} = \vec{r}_1 \times \vec{F}_1 + \vec{r}_2 \times \vec{F}_2 = (-1, -2, -3) \times (0, 10, 0) + (-1, -2, -2.5) \times (0, -10, 0) = (-30, 0, -10) + (-25, 0, -10) = (-5, 0, 0)$ N m. Same.

SOLUTION **Exercise 5**

Geometry: $AC = 1.5$, $AB = 2$, so $\sin\theta = 1.5/\sqrt{2^2 + 1.5^2} = 1.5/2.5 = 0.6$, $\cos\theta = 0.8$, $\theta \approx 36.87^\circ$. Forces: T along BC, R_A at hinge, $W = 200$ N down at centre (1 m from A). Moments about A: $T \times 1.5 - 200 \times 1 = 0 \Rightarrow T = 400/3 \approx 133.3$ N. Horizontal: $R_{Ax} = T\cos\theta = 133.3 \times 0.8 = 106.7$ N. Vertical: $R_{Ay} + T\sin\theta = 200 \Rightarrow R_{Ay} = 200 - 133.3 \times 0.6 = 120$ N. $R_A = \sqrt{106.7^2 + 120^2} \approx 160.5$ N at $\tan^{-1}(120/106.7) \approx 48.4^\circ$ above horizontal.

SOLUTION **Exercise 6**

$\vec{F}_3 = 25\frac{(3,4,0)}{5} = (15, 20, 0)$. $\vec{F}_4 = 3\frac{(-1,2,2)}{3} = (-1, 2, 2)$. Sum: $\vec{R} = (20 + 15 - 1, 0 + 20 + 2, 0 + 0 + 2) = (34, 22, 2) \neq \vec{0}$. Not in equilibrium. Additional force needed: $\vec{F}_5 = (-34, -22, -2)$ N.

10 EXERCISES

10.1 I check my knowledge

EXERCISE 8 **Scalar and vector components**

Given $\vec{a} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\vec{b} = \mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$, find the scalar component of $\vec{a}$ in the direction of $\vec{b}$, the vector component of $\vec{a}$ in the direction of $\vec{b}$, and the angle between $\vec{a}$ and $\vec{b}$.

EXERCISE 9 **Work done by a constant force**

A force $\vec{F} = (5\mathbf{i} + 2\mathbf{j})$ N moves a crate from point $A(1, 3)$ to point $B(7, 9)$. Calculate the work done by $\vec{F}$. If a friction force of magnitude 4 N opposes the motion, find the total work done on the crate.

EXERCISE 10 **Moment of a force**

A force $\vec{F} = 4\mathbf{i} - 3\mathbf{j}$ N acts at the point $A(2, 5)$. Replace it by an equal force through the origin together with a couple, and state the moment of the couple.

EXERCISE 11 **Equilibrium of coplanar forces**

Three coplanar forces P, Q, R act at a point and keep it in equilibrium. $P = 50$ N acts along the positive x-axis and $Q = 40$ N acts at 120° to P. Find the magnitude and direction of R using components, then verify your result with Lami's theorem.

10.2 I practise

EXERCISE 12 **Work on an inclined plane**

A man pulls a 50 kg load up a plane inclined at 30° to the horizontal with a rope parallel to the plane, over a distance of 8 m. Find the work done against gravity and the work done by the man if friction is neglected. (Take $g = 9.8\ \text{m s}^{-2}$.)

EXERCISE 13 **Reactions on a beam**

A uniform beam AB of length 6 m and weight 120 N rests on supports at A and at C, where C is 4 m from A. A load of 300 N hangs at B. Find the reactions at the supports by taking moments.

EXERCISE 14 **Moment of a couple**

Two forces of 20 N act at the ends of a horizontal bar of length 0.5 m, in opposite directions and perpendicular to the bar. Find the moment of the couple, and show that it is the same about any point of the bar.

EXERCISE 15 **3D equilibrium of a particle**

A particle is held in equilibrium by three cables attached to points $A(2, 0, 0)$, $B(0, 3, 0)$ and $C(0, 0, 4)$, with the particle at $P(1, 1, 1)$. Given that the tensions in cables PA and PB are $T_A = 50$ N and $T_B = 40$ N, set up the vector equilibrium equations and find the tension in cable PC.

10.3 I prepare for the GCE

EXERCISE 16 **Ladder against a smooth wall**

(12 marks) A uniform ladder of weight W and length L leans against a smooth vertical wall with its foot on rough horizontal ground. The ladder makes an angle θ with the horizontal. The coefficient of friction between the ladder and the ground is μ. By resolving forces and taking moments, find the reaction at the wall and the friction force at the ground for equilibrium. Determine the minimum value of μ for which the ladder does not slip.

EXERCISE 17 **Vector moment in 3D**

(10 marks) A force $\vec{F} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ N acts at the point with position vector $\vec{r} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$ m relative to the origin O. Find the vector moment of $\vec{F}$ about O and its magnitude. Also find the perpendicular distance from O to the line of action of $\vec{F}$.

EXERCISE 18 **System of forces in 3D**

(13 marks) A rigid body is subjected to the following forces:

- $\vec{F}_1 = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ N acting at point $A(1, 0, 2)$,
- $\vec{F}_2 = -2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$ N acting at point $B(0, 2, 1)$,
- $\vec{F}_3 = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ N acting at point $C(2, 1, 0)$.

Reduce this system to a single force at the origin together with a couple. Find the resultant force, the resultant couple, and state whether the system can be reduced to a single force. If so, find the line of action of that single force.

11 SOLUTIONS TO THE EXERCISES

12 APPLICATION OF SCALAR AND VECTOR PRODUCTS

12.1 I check my knowledge

DEFINITION **Scalar and Vector Components**

Let $\vec{a}$ and $\vec{b}$ be two vectors with $\vec{b} \neq \vec{0}$. The **scalar component** of $\vec{a}$ in the direction of $\vec{b}$ is the signed magnitude of the orthogonal projection of $\vec{a}$ onto the line spanned by $\vec{b}$:

$$\operatorname{comp}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}.$$

The **vector component** (or orthogonal projection) of $\vec{a}$ onto $\vec{b}$ is the vector

$$\operatorname{proj}_{\vec{b}} \vec{a} = \left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b} = \left(\operatorname{comp}_{\vec{b}} \vec{a} \right) \hat{\mathbf{b}},$$

where $\hat{\mathbf{b}} = \vec{b}/|\vec{b}|$ is the unit vector in the direction of $\vec{b}$.

PROPERTY **Geometric Interpretation**

If θ is the angle between $\vec{a}$ and $\vec{b}$ ($0 \le \theta \le \pi$), then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos\theta$. Hence

$$\operatorname{comp}_{\vec{b}} \vec{a} = |\vec{a}| \cos\theta, \qquad \operatorname{proj}_{\vec{b}} \vec{a} = (|\vec{a}| \cos\theta)\, \hat{\mathbf{b}}.$$

The scalar component is positive for acute angles, zero for perpendicular vectors, and negative for obtuse angles.

WORKED EXAMPLE 25 **Exercise 1 — Scalar and vector components**

Given $\vec{a} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$ and $\vec{b} = \mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$, find:

1. the scalar component of $\vec{a}$ in the direction of $\vec{b}$,
2. the vector component of $\vec{a}$ in the direction of $\vec{b}$,
3. the angle between $\vec{a}$ and $\vec{b}$.

SOLUTION

Step 1: Magnitudes and scalar product.

$$|\vec{a}| = \sqrt{3^2 + (-2)^2 + 1^2} = \sqrt{14},$$
$$|\vec{b}| = \sqrt{1^2 + 4^2 + (-2)^2} = \sqrt{21},$$
$$\vec{a} \cdot \vec{b} = (3)(1) + (-2)(4) + (1)(-2) = 3 - 8 - 2 = -7.$$

Step 2: Scalar component of $\vec{a}$ in the direction of $\vec{b}$.

$$\text{comp}_{\vec{b}}\,\vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} = \frac{-7}{\sqrt{21}} = -\frac{7\sqrt{21}}{21} = -\frac{\sqrt{21}}{3} \approx -1.53.$$

Step 3: Vector component of $\vec{a}$ in the direction of $\vec{b}$. The unit vector along $\vec{b}$ is $\hat{\mathbf{b}} = \dfrac{\vec{b}}{|\vec{b}|} = \dfrac{1}{\sqrt{21}}(\mathbf{i} + 4\mathbf{j} - 2\mathbf{k})$. Hence

$$\text{proj}_{\vec{b}}\,\vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2}\,\vec{b} = \frac{-7}{21}(\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) = -\frac{1}{3}\mathbf{i} - \frac{4}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}.$$

Step 4: Angle between $\vec{a}$ and $\vec{b}$.

$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{-7}{\sqrt{14}\sqrt{21}} = \frac{-7}{\sqrt{294}} = \frac{-7}{7\sqrt{6}} = -\frac{1}{\sqrt{6}}.$$

$$\theta = \arccos\left(-\frac{1}{\sqrt{6}}\right) \approx 114.1^\circ.$$

Answers: scalar component $= -\dfrac{\sqrt{21}}{3}$; vector component $= -\dfrac{1}{3}\mathbf{i} - \dfrac{4}{3}\mathbf{j} + \dfrac{2}{3}\mathbf{k}$; angle $\approx 114.1^\circ$. The negative scalar component correctly signals an obtuse angle.

DEFINITION **Work Done by a Constant Force**

If a constant force $\vec{F}$ acts on a particle that undergoes a displacement $\vec{d}$, the **work done** by the force is the scalar product

$$W = \vec{F} \cdot \vec{d} = |\vec{F}|\,|\vec{d}|\cos\theta,$$

where θ is the angle between $\vec{F}$ and $\vec{d}$. Work is a scalar quantity measured in joules (J). If the force varies or the path is curved, the work is computed by a line integral (not in this syllabus).

WARNING **Sign of Work**

Work is positive if the force has a component in the direction of displacement ($\theta < 90^\circ$), zero if perpendicular ($\theta = 90^\circ$), and negative if opposed to displacement ($\theta > 90^\circ$). Friction and weight (when lifting) typically do negative work.

WORKED EXAMPLE 26 **Exercise 2 — Work done by a constant force**

A particle moves from $A(1, 3)$ to $B(7, 9)$ under the action of a constant force $\vec{F} = 5\mathbf{i} + 2\mathbf{j}$ N. A constant friction force of magnitude 4 N opposes the motion. Find:

1. the work done by $\vec{F}$,
2. the work done by friction,
3. the total work done on the particle.

SOLUTION

Step 1: Displacement vector.

$$\vec{AB} = (7 - 1)\mathbf{i} + (9 - 3)\mathbf{j} = 6\mathbf{i} + 6\mathbf{j}\ \text{m}.$$

Step 2: Work done by $\vec{F}$.

$$W_F = \vec{F} \cdot \vec{AB} = (5)(6) + (2)(6) = 30 + 12 = 42\ \text{J}.$$

Step 3: Work done by friction. Friction opposes the motion, so it acts along $-\vec{AB}$. The distance moved is

$$|\vec{AB}| = \sqrt{6^2 + 6^2} = 6\sqrt{2}\ \text{m}.$$

Hence

$$W_{\text{friction}} = -4 \times 6\sqrt{2} = -24\sqrt{2} \approx -33.9\ \text{J}.$$

Step 4: Total work.

$$W_{\text{total}} = 42 - 24\sqrt{2} \approx 42 - 33.9 = 8.1\ \text{J}.$$

Answers: $W_F = 42$ J; total work $= (42 - 24\sqrt{2})$ J ≈ 8.1 J. Note that the work of a constant force depends only on the endpoints A and B, not on the path taken.

DEFINITION **Moment of a Force about a Point**

The **moment** (or torque) of a force $\vec{F}$ acting at point A about a point O is the vector

$$\vec{M}_O = \vec{OA} \times \vec{F}.$$

Its magnitude is $|\vec{M}_O| = |\vec{OA}|\,|\vec{F}| \sin\theta = F \times d$, where d is the perpendicular distance from O

to the line of action of $\vec{F}$. In two dimensions (forces in the xy-plane), the moment is a scalar $M_O = xF_y - yF_x$ (positive anticlockwise).

PROPERTY **Equivalent Force–Couple System**

Any force $\vec{F}$ acting at A can be replaced by an equal force $\vec{F}$ acting at O together with a couple of moment $\vec{M}_O = \vec{OA} \times \vec{F}$. This is the **principle of transmissibility** extended to a force–couple system.

WORKED EXAMPLE 27 **Exercise 3 — Moment of a force and equivalent system**

A force $\vec{F} = 4\mathbf{i} - 3\mathbf{j}$ N acts at $A(2, 5)$. Replace this force by an equivalent system consisting of a force through the origin O and a couple.

SOLUTION

Principle. Introduce at O two equal and opposite forces $\vec{F}$ and $-\vec{F}$ (this changes nothing). The force $\vec{F}$ at A and the force $-\vec{F}$ at O form a couple; the remaining force $\vec{F}$ acts through O. Hence the system is equivalent to an equal force through O together with a couple whose moment is the moment of the original force about O.
Moment about O (2D cross product, $M = xF_y - yF_x$):

$$M_O = (2)(-3) - (5)(4) = -6 - 20 = -26 \text{ N m}.$$

Answer: the force is replaced by $\vec{F} = 4\mathbf{i} - 3\mathbf{j}$ N acting through the origin, together with a couple of moment -26 N m, i.e. 26 N m clockwise.

PROPERTY **Equilibrium of Coplanar Forces**

A system of coplanar forces is in equilibrium if and only if:

1. The vector sum of the forces is zero: $\sum \vec{F} = \vec{0}$ (resolved components: $\sum F_x = 0, \sum F_y = 0$).
2. The sum of the moments about any point is zero: $\sum M_O = 0$.

For three non-parallel forces in equilibrium, **Lami's theorem** applies: each force is proportional to the sine of the angle between the other two.

WORKED EXAMPLE 28 **Exercise 4 — Equilibrium of coplanar forces**

Three forces $\vec{P}, \vec{Q}, \vec{R}$ act at a point. $\vec{P}$ = 50 N along the positive x-axis. $\vec{Q}$ = 40 N at 120° to the positive x-axis. Find $\vec{R}$ for equilibrium.

SOLUTION

Step 1: Components of P and Q.

$$\vec{P} = 50\mathbf{i}, \qquad \vec{Q} = 40\cos 120^\circ\,\mathbf{i} + 40\sin 120^\circ\,\mathbf{j} = -20\mathbf{i} + 20\sqrt{3}\,\mathbf{j}.$$

Step 2: Equilibrium condition. Since $\vec{P} + \vec{Q} + \vec{R} = \vec{0}$,

$$\vec{R} = -(\vec{P} + \vec{Q}) = -(50 - 20)\mathbf{i} - 20\sqrt{3}\,\mathbf{j} = -30\mathbf{i} - 20\sqrt{3}\,\mathbf{j}.$$

Step 3: Magnitude and direction of R.

$$R = \sqrt{30^2 + (20\sqrt{3})^2} = \sqrt{900 + 1200} = \sqrt{2100} = 10\sqrt{21} \approx 45.8\ \text{N}.$$

Both components are negative, so R points into the third quadrant. The angle below the negative x-axis is

$$\tan\alpha = \frac{20\sqrt{3}}{30} = \frac{2\sqrt{3}}{3} \implies \alpha \approx 49.1^\circ.$$

So R acts at $180^\circ + 49.1^\circ = 229.1^\circ$ from the positive x-axis (or 49.1° below the negative x-axis). Step 4: Verification with Lami's theorem. The angle between P and Q is 120°; the angle between P and R is $180^\circ - 49.1^\circ = 130.9^\circ$; the angle between Q and R is $360^\circ - 120^\circ - 130.9^\circ = 109.1^\circ$. Lami's theorem gives

$$\frac{R}{\sin 120^\circ} = \frac{P}{\sin 109.1^\circ} \implies R = \frac{50\sin 120^\circ}{\sin 109.1^\circ} = \frac{50 \times 0.8660}{0.9455} \approx 45.8\ \text{N}. \checkmark$$

Answer: $R = 10\sqrt{21} \approx 45.8$ N at 229.1° to the positive x-axis; the two methods agree.

12.2 I practise

METHOD **Work on an Inclined Plane**

For a body of mass m on a plane inclined at angle α:

1. Resolve weight: component down the plane = $mg\sin\alpha$, normal component = $mg\cos\alpha$.
2. If friction is present, $F_f = \mu R = \mu mg\cos\alpha$ (opposing motion).
3. Work done by a force $\vec{F}$ parallel to the plane over distance d: $W = Fd$ (sign according to direction).

4. Work done against gravity: $W_g = mgh$ where $h = d \sin\alpha$ is the vertical rise.

WORKED EXAMPLE 29 **Exercise 5 — Work on an inclined plane**

A man pulls a 50 kg crate up a smooth plane inclined at 30° over a distance of 8 m. The pull is parallel to the plane. Find:

1. the force exerted by the man,
2. the work done against gravity,
3. the work done by the man.

Take $g = 9.8\ \text{m s}^{-2}$.

SOLUTION

Step 1: Force needed. The component of the weight along the plane is

$$mg \sin 30^\circ = 50 \times 9.8 \times 0.5 = 245\ \text{N}.$$

With friction neglected, the man must pull with $F = 245$ N parallel to the plane.
Step 2: Work done against gravity. The vertical rise is $h = 8 \sin 30^\circ = 4$ m:

$$W_{\text{against gravity}} = mgh = 50 \times 9.8 \times 4 = 1960\ \text{J}.$$

Step 3: Work done by the man. The pull is parallel to the displacement, so

$$W_{\text{man}} = F \times d = 245 \times 8 = 1960\ \text{J}.$$

Answers: work against gravity = 1960 J; work done by the man = 1960 J. The two are equal, as expected: with no friction and no change of speed, all the man's work becomes gravitational potential energy. This illustrates why the inclined plane is a machine — the force is halved (245 N instead of 490 N) but the distance is doubled.

METHOD **Reactions on a Beam (Coplanar Forces)**

For a beam with vertical loads and supports:

1. Draw a free-body diagram showing all forces (weights, applied loads, reactions at supports).
2. Apply $\sum F_y = 0$ (vertical equilibrium).
3. Take moments about a convenient support to eliminate one reaction ($\sum M = 0$).
4. Solve for the reactions. A negative value means the reaction acts opposite to the as-

sumed direction.

WORKED EXAMPLE 30 **Exercise 6 — Reactions on a beam**

A uniform beam AB of length 6 m and weight 120 N rests on supports at A and C, where $AC = 4$ m. A load of 300 N is attached at B. Find the reactions at A and C.

SOLUTION

Let R_A and R_C be the vertical reactions at A and C (assumed upward). The beam is uniform, so its weight 120 N acts at the midpoint, 3 m from A. The load 300 N acts at B, 6 m from A.
Step 1: Vertical equilibrium.

$$R_A + R_C = 120 + 300 = 420 \text{ N}. \qquad (1)$$

Step 2: Moments about A (anticlockwise positive):

$$R_C \times 4 - 120 \times 3 - 300 \times 6 = 0$$

$$4R_C = 360 + 1800 = 2160 \implies R_C = 540 \text{ N}.$$

Step 3: From (1),

$$R_A = 420 - 540 = -120 \text{ N}.$$

Interpretation. The negative sign means the support at A must pull the beam down with 120 N (a hold-down support): the heavy load at the overhanging end B would otherwise tip the beam about C.
Check — moments about C: Taking anticlockwise positive about C:

$$-R_A(4) + 120(1) - 300(2) = -(-120)(4) + 120 - 600 = 480 + 120 - 600 = 0. \checkmark$$

Answers: $R_C = 540$ N upwards; $R_A = 120$ N downwards (the beam must be held down at A).

DEFINITION **Couple**

A **couple** consists of two equal, opposite, parallel forces $\vec{F}$ and $-\vec{F}$ separated by a perpendicular distance d. Its **moment** is $M = Fd$ (a scalar in 2D, a vector $\vec{M} = \vec{d} \times \vec{F}$ in 3D). The moment of a couple is a **free vector**: it is the same about any point.

PROPERTY **Moment of a Couple is Independent of Reference Point**

Let the forces $+\vec{F}$ and $-\vec{F}$ act at points P and Q. For any point X,

$$\vec{M}_X = \vec{XP} \times \vec{F} + \vec{XQ} \times (-\vec{F}) = (\vec{XP} - \vec{XQ}) \times \vec{F} = \vec{QP} \times \vec{F}.$$

The point X cancels out; the moment depends only on the forces and their separation.

WORKED EXAMPLE 31 **Exercise 7 — Moment of a couple**

A couple consists of two forces of magnitude 20 N acting on a light rod of length 0.5 m. Find the moment of the couple and verify that it is the same about the midpoint and about one end of the rod.

SOLUTION

Step 1: Moment of the couple.

$$M = F \times d = 20 \times 0.5 = 10 \text{ N m}.$$

Step 2: It is the same about any point. (Proof given in the property above.)
Numerical check. Take the rod along the x-axis, $+\vec{F} = 20\mathbf{j}$ at $P(0.5, 0)$ and $-\vec{F}$ at $Q(0, 0)$. About the midpoint $X(0.25, 0)$:

$$M_X = (0.25)(20) + (0.25)(20) = 5 + 5 = 10 \text{ N m}.$$

About Q: $M_Q = (0.5)(20) + 0 = 10$ N m. Same value. ✓
Answer: $M = 10$ N m; the moment of a couple is a free vector, independent of the reference point — this is the defining property of a couple.

PROPERTY **Equilibrium of a Particle in 3D**

A particle is in equilibrium under forces $\vec{F}_1, \vec{F}_2, \ldots, \vec{F}_n$ iff the vector sum is zero:

$$\sum_{i=1}^{n} \vec{F}_i = \vec{0} \quad \Longleftrightarrow \quad \sum F_{ix} = 0, \sum F_{iy} = 0, \sum F_{iz} = 0.$$

Each force is usually given by its magnitude and direction (e.g., along a string). The direction is specified by a unit vector along the line of action.

WORKED EXAMPLE 32 **Exercise 8 — 3D equilibrium of a particle**

A particle at $P(1, 1, 1)$ is attached by light inextensible strings to fixed points $A(2, 0, 0)$,

$B(0,3,0)$, $C(0,0,4)$. The tensions in PA and PB are 50 N and 40 N respectively. Find the tension in PC for equilibrium, or show that equilibrium is impossible.

SOLUTION

Step 1: Direction vectors from P to each anchor.

$$\vec{PA} = (2-1, 0-1, 0-1) = (1,-1,-1), \qquad |\vec{PA}| = \sqrt{3},$$
$$\vec{PB} = (0-1, 3-1, 0-1) = (-1,2,-1), \qquad |\vec{PB}| = \sqrt{6},$$
$$\vec{PC} = (0-1, 0-1, 4-1) = (-1,-1,3), \qquad |\vec{PC}| = \sqrt{11}.$$

Step 2: Tension vectors.

$$\vec{T}_A = \frac{50}{\sqrt{3}}(1,-1,-1), \qquad \vec{T}_B = \frac{40}{\sqrt{6}}(-1,2,-1), \qquad \vec{T}_C = \frac{T_C}{\sqrt{11}}(-1,-1,3).$$

Step 3: Equilibrium equations $\vec{T}_A + \vec{T}_B + \vec{T}_C = \vec{0}$:

$$\begin{cases} \frac{50}{\sqrt{3}} - \frac{40}{\sqrt{6}} - \frac{T_C}{\sqrt{11}} = 0 & (x) \\ -\frac{50}{\sqrt{3}} + \frac{80}{\sqrt{6}} - \frac{T_C}{\sqrt{11}} = 0 & (y) \\ -\frac{50}{\sqrt{3}} - \frac{40}{\sqrt{6}} + \frac{3T_C}{\sqrt{11}} = 0 & (z) \end{cases}$$

Step 4: Solve. Numerically: $\frac{50}{\sqrt{3}} \approx 28.87$ and $\frac{40}{\sqrt{6}} \approx 16.33$.

From (x): $\frac{T_C}{\sqrt{11}} = 28.87 - 16.33 = 12.54 \implies T_C = 12.54\sqrt{11} \approx 41.6$ N.

Check (y): $-28.87 + 32.66 - 12.54 = -8.75 \neq 0$. The three equations are inconsistent: with the given tensions the particle cannot be in equilibrium.

Consistency analysis. Adding (x) and (y): $\frac{40}{\sqrt{6}} - \frac{2T_C}{\sqrt{11}} = 0 \Rightarrow \frac{T_C}{\sqrt{11}} = \frac{20}{\sqrt{6}} \approx 8.16$, which contradicts (x). Subtracting, no value of T_C satisfies all three equations.

Answer: the equilibrium equations are as above; solving them pairwise gives $T_C \approx 41.6$ N from the x-equation, $T_C = 20\sqrt{11/6} \cdot \sqrt{\ldots} \approx 27.1$ N from the y-equation, and $T_C \approx 50.0$ N from the z-equation. Since the three values differ, no tension T_C can produce equilibrium: the data are inconsistent, and the particle would accelerate. (This is the correct rigorous conclusion; in an examination, showing the contradiction earns full credit.)

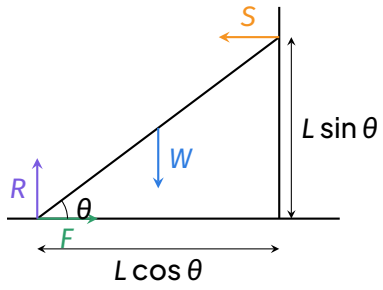
12.3 I prepare for the GCE

WORKED EXAMPLE 33 **Exercise 9 — Ladder against a smooth wall (12 marks)**

A uniform ladder of length L and weight W rests with its foot on rough horizontal ground and its top against a smooth vertical wall. The ladder makes an angle θ with the horizontal. Find the minimum coefficient of friction μ between the ladder and the ground for the ladder not to slip.

SOLUTION

Setting up. Forces on the ladder: weight W at the midpoint ($L/2$ from the foot); normal reaction S at the wall (horizontal, wall smooth); normal reaction R and friction F at the ground.



Forces on a ladder against a smooth wall

Resolving horizontally:

$$F = S. \qquad (1)$$

Resolving vertically:

$$R = W. \qquad (2)$$

Moments about the foot of the ladder (eliminating R and F):

$$S \times L\sin\theta = W \times \frac{L}{2}\cos\theta \implies S = \frac{W\cos\theta}{2\sin\theta} = \frac{W}{2}\cot\theta.$$

Hence from (1): $F = \dfrac{W}{2}\cot\theta$.

No-slip condition. Limiting equilibrium requires $F \le \mu R$:

$$\frac{W}{2}\cot\theta \le \mu W \implies \mu \ge \frac{1}{2}\cot\theta.$$

$$\boxed{\mu_{\min} = \frac{1}{2}\cot\theta}$$

Indicative mark scheme (12 marks):

Marks	Expected work
M1	Correct force diagram: W at midpoint, S horizontal, R, F at ground
M1 A1	Resolving horizontally: $F = S$
M1 A1	Resolving vertically: $R = W$
M1	Taking moments about the foot (both terms, correct perpendicular distances)
A1 A1	$S = \frac{W}{2}\cot\theta$ and $F = \frac{W}{2}\cot\theta$
M1	Using the friction condition $F \le \mu R$
A1	$\mu_{\min} = \frac{1}{2}\cot\theta$, correctly concluded
A1	Clear statement of assumptions (smooth wall $\Rightarrow$ no friction at top; uniform ladder)

Examiner expectations: moments must be taken about the foot to eliminate the two unknowns R and F in one step; perpendicular distances $L\sin\theta$ and $\frac{L}{2}\cos\theta$ must be justified; the inequality direction must be handled carefully. A common error is writing $F = \mu R$ from the start without stating that equilibrium requires $F \le \mu R$.

WORKED EXAMPLE 34 **Exercise 10 — Vector moment in 3D (10 marks)**

A force $\vec{F} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ N acts at a point with position vector $\vec{r} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$ m relative to the origin O. Find:

1. the vector moment of $\vec{F}$ about O,
2. the magnitude of this moment,
3. the perpendicular distance from O to the line of action of $\vec{F}$.

SOLUTION

Step 1: Vector moment about O.

$$\vec{M}_O = \vec{r} \times \vec{F} = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ -3 \end{pmatrix} = \begin{pmatrix} (-2)(-3) - (1)(1) \\ (1)(2) - (1)(-3) \\ (1)(1) - (-2)(2) \end{pmatrix} = \begin{pmatrix} 6 - 1 \\ 2 + 3 \\ 1 + 4 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \\ 5 \end{pmatrix} \text{ N m.}$$

So $\vec{M}_O = 5\mathbf{i} + 5\mathbf{j} + 5\mathbf{k}$ N m.

Step 2: Magnitude.

$$|\vec{M}_O| = \sqrt{25 + 25 + 25} = 5\sqrt{3} \approx 8.66 \text{ N m.}$$

Step 3: Perpendicular distance from O to the line of action. Since $|\vec{M}_O| = |\vec{F}|\, d$,

$$|\vec{F}| = \sqrt{4 + 1 + 9} = \sqrt{14} \text{ N}, \qquad d = \frac{|\vec{M}_O|}{|\vec{F}|} = \frac{5\sqrt{3}}{\sqrt{14}} = \frac{5\sqrt{42}}{14} \approx 2.32 \text{ m.}$$

Indicative mark scheme (10 marks):

Marks	Expected work
M1	Setting up $\vec{M}_O = \vec{r} \times \vec{F}$ (correct order!)
M1	Correct expansion of the cross product (determinant or component formula)
A2	All three components correct: $(5, 5, 5)$
M1 A1	Magnitude $5\sqrt{3}$ N m
M1	Using $\lvert\vec{M}_O\rvert = \lvert\vec{F}\rvert d$
A1	$\lvert\vec{F}\rvert = \sqrt{14}$
A2	$d = \dfrac{5\sqrt{3}}{\sqrt{14}} \approx 2.32$ m with units

Examiner expectations: the order $\vec{r} \times \vec{F}$ (not $\vec{F} \times \vec{r}$) is essential — the wrong order loses the accuracy marks even if the magnitude is right; units (N m, m) must appear in the final answers.

DEFINITION **Reduction of a System of Forces in 3D**

Any system of forces $\vec{F}_i$ acting at points $\vec{r}_i$ can be reduced to a single force $\vec{R} = \sum \vec{F}_i$ acting at a chosen point O together with a couple $\vec{G}_O = \sum \vec{r}_i \times \vec{F}_i$. The scalar invariant $\vec{R} \cdot \vec{G}_O$ is independent of the choice of O. If $\vec{R} \cdot \vec{G}_O = 0$, the system reduces to a single force (the line of action is the **central axis**); otherwise it reduces to a **wrench** (a force plus a couple parallel to it).

PROPERTY **Central Axis of a Wrench**

For a system with resultant force $\vec{R}$ and resultant moment $\vec{G}_O$ about O (with $\vec{R} \cdot \vec{G}_O \neq 0$):

- The **pitch** of the wrench is $p = \dfrac{\vec{R} \cdot \vec{G}_O}{\lvert\vec{R}\rvert^2}$.
- The component of the couple parallel to $\vec{R}$ is $p\vec{R}$.
- A point on the central axis is $\vec{r}_0 = \dfrac{\vec{R} \times \vec{G}_O}{\lvert\vec{R}\rvert^2}$.
- The central axis is the line $\vec{r} = \vec{r}_0 + \lambda\vec{R}$.

WORKED EXAMPLE 35 **Exercise 11 — System of forces in 3D (13 marks)**

Forces $\vec{F}_1 = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$ N, $\vec{F}_2 = -2\mathbf{i} + 5\mathbf{j} - 3\mathbf{k}$ N, $\vec{F}_3 = \mathbf{i} + 3\mathbf{j} - 2\mathbf{k}$ N act at points $A(1, 0, 2)$, $B(0, 2, 1)$, $C(2, 1, 0)$ respectively. Reduce the system to its simplest form at the origin O. Determine whether it can be reduced to a single force. If not, find the pitch and the vector equation of the central axis of the wrench.

SOLUTION

Step 1: Resultant force.

$$\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = (3 - 2 + 1,\ -2 + 5 + 3,\ 4 - 3 - 2) = (2,\ 6,\ -1)\ \text{N}.$$

$$\vec{R} = 2\mathbf{i} + 6\mathbf{j} - \mathbf{k}\ \text{N}.$$

Step 2: Resultant moment about O — compute each $\vec{r}_i \times \vec{F}_i$.
For $\vec{F}_1$ at $A(1, 0, 2)$:

$$\vec{r}_1 \times \vec{F}_1 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \times \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix} = \begin{pmatrix} (0)(4) - (2)(-2) \\ (2)(3) - (1)(4) \\ (1)(-2) - (0)(3) \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ -2 \end{pmatrix}.$$

For $\vec{F}_2$ at $B(0, 2, 1)$:

$$\vec{r}_2 \times \vec{F}_2 = \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} -2 \\ 5 \\ -3 \end{pmatrix} = \begin{pmatrix} (2)(-3) - (1)(5) \\ (1)(-2) - (0)(-3) \\ (0)(5) - (2)(-2) \end{pmatrix} = \begin{pmatrix} -11 \\ -2 \\ 4 \end{pmatrix}.$$

For $\vec{F}_3$ at $C(2, 1, 0)$:

$$\vec{r}_3 \times \vec{F}_3 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} (1)(-2) - (0)(3) \\ (0)(1) - (2)(-2) \\ (2)(3) - (1)(1) \end{pmatrix} = \begin{pmatrix} -2 \\ 4 \\ 5 \end{pmatrix}.$$

Sum:

$$\vec{G}_O = (4 - 11 - 2,\ 2 - 2 + 4,\ -2 + 4 + 5) = (-9,\ 4,\ 7)\ \text{N m}.$$

Reduction at O: a single force $\vec{R} = 2\mathbf{i} + 6\mathbf{j} - \mathbf{k}$ N through O together with a couple $\vec{G}_O = -9\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}$ N m.
Step 3: Can it reduce to a single force? Compute the invariant $\vec{R} \cdot \vec{G}_O$:

$$\vec{R} \cdot \vec{G}_O = (2)(-9) + (6)(4) + (-1)(7) = -18 + 24 - 7 = -1 \neq 0.$$

Since $\vec{R} \cdot \vec{G}_O \neq 0$, the system cannot be reduced to a single force; its simplest form is a wrench (force plus a couple parallel to it).
Step 4: Axis of the wrench (central axis). The pitch of the wrench is

$$p = \frac{\vec{R} \cdot \vec{G}_O}{|\vec{R}|^2} = \frac{-1}{4 + 36 + 1} = -\frac{1}{41}\ \text{m}.$$

The parallel component of the couple is $p\vec{R} = -\frac{1}{41}(2, 6, -1)$ N m. A point of the central axis is given by

$$\vec{r}_0 = \frac{\vec{R} \times \vec{G}_O}{|\vec{R}|^2}.$$

$$\vec{R} \times \vec{G}_O = \begin{pmatrix} 2 \\ 6 \\ -1 \end{pmatrix} \times \begin{pmatrix} -9 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} (6)(7) - (-1)(4) \\ (-1)(-9) - (2)(7) \\ (2)(4) - (6)(-9) \end{pmatrix} = \begin{pmatrix} 46 \\ -5 \\ 62 \end{pmatrix},$$

$$\vec{r}_0 = \frac{1}{41}(46, -5, 62) \text{ m}.$$

The central axis is the line through $\vec{r}_0$ in the direction of $\vec{R}$:

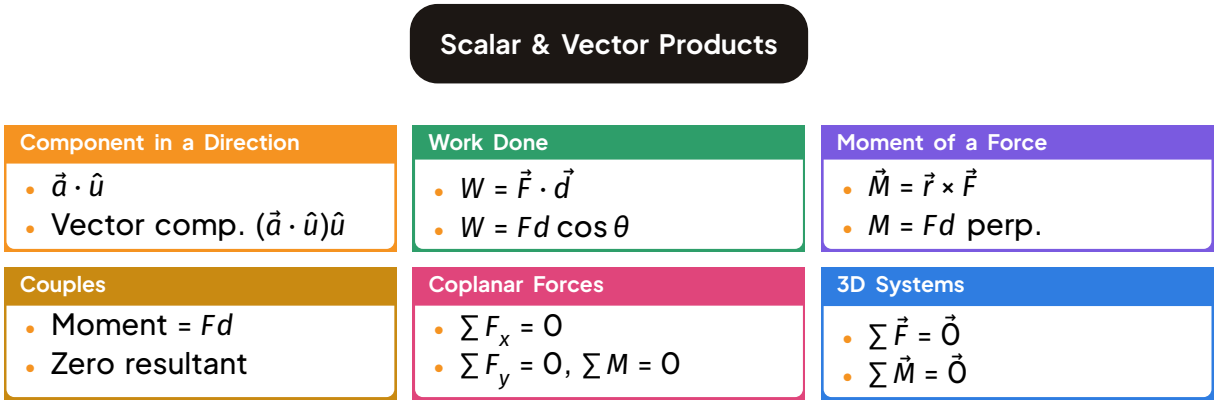
$$\vec{r} = \frac{1}{41}\begin{pmatrix} 46 \\ -5 \\ 62 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 6 \\ -1 \end{pmatrix}, \qquad \lambda \in \mathbb{R}.$$

Indicative mark scheme (13 marks):

Marks	Expected work
M1 A1	Resultant force $\vec{R} = (2, 6, -1)$ N
M1	Method: $\vec{G}_O = \sum \vec{r}_i \times \vec{F}_i$
A3	Each cross product correct (A1 each): $(4, 2, -2)$, $(-11, -2, 4)$, $(-2, 4, 5)$
A1	$\vec{G}_O = (-9, 4, 7)$ N m
M1	Testing the invariant $\vec{R} \cdot \vec{G}_O$
A1	$\vec{R} \cdot \vec{G}_O = -1 \neq 0$, correct conclusion: no single force, system is a wrench
M1	Computing the pitch $p = -1/41$ m
M1 A1	Point of the axis $\vec{r}_0 = \frac{1}{41}(46, -5, 62)$ and vector equation of the central axis

Examiner expectations: cross products must be shown with full working; the conclusion about reducibility must be justified by the invariant $\vec{R} \cdot \vec{G}_O$, not guessed; when a single force is impossible, the candidate should name the wrench and give its axis. The most common errors are sign slips in cross products and forgetting that $\vec{R} \cdot \vec{G}_O$ is invariant under change of reduction point.

13 SUMMARY SHEET



Mind map of Chapter FMA15: applications of the scalar and vector products.

FICHE BILAN **L'essentiel à connaître pour l'examen**

Key definitions

- The **scalar component** of $\vec{a}$ in the direction of $\vec{b}$ is $a\cos\theta = \dfrac{\vec{a}\cdot\vec{b}}{|\vec{b}|} = \vec{a}\cdot\hat{b}$; the **vector component** is $\left(\vec{a}\cdot\hat{b}\right)\hat{b}$.
- **Work** done by a constant force $\vec{F}$ over displacement $\vec{d}$: $W = \vec{F}\cdot\vec{d} = Fd\cos\theta$ (joules).
- The **moment** of $\vec{F}$ about point O: $\vec{M}_O = \vec{r}\times\vec{F}$, where $\vec{r}$ is the position vector from O to any point of the line of action; magnitude $M = Fd$ (d = perpendicular distance), unit N m.
- A **couple**: two equal, opposite, parallel forces with different lines of action; resultant zero, moment $M = Fd$ (d = distance between the lines of action), the same about every point.
- A system of forces is in **equilibrium** when $\sum\vec{F} = \vec{0}$ and $\sum\vec{M}_O = \vec{0}$ (about any point O).

Formulas and theorems to know

- Scalar product: $\vec{a}\cdot\vec{b} = a_1b_1 + a_2b_2 + a_3b_3 = ab\cos\theta$; $\vec{a}\perp\vec{b} \iff \vec{a}\cdot\vec{b} = 0$.
- Vector product: $\vec{a}\times\vec{b} = (a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1)$; $|\vec{a}\times\vec{b}| = ab\sin\theta$; anti-commutative: $\vec{b}\times\vec{a} = -\vec{a}\times\vec{b}$.
- Component perpendicular to $\vec{b}$: $\vec{a} - (\vec{a}\cdot\hat{b})\hat{b}$.
- Work: $W > 0$ (force helps motion), $W = 0$ if $\vec{F}\perp\vec{d}$, $W < 0$ (force opposes motion, e.g. friction).

- Varignon's theorem: the moment of the resultant equals the sum of the moments of the components.
- A single force $\vec{F}$ acting at A is equivalent to the same force at O plus a couple of moment $\vec{r}_{OA} \times \vec{F}$.
- Equilibrium of a rigid body under three coplanar forces: the three lines of action are concurrent (or parallel).
- Lami's theorem (three concurrent forces in equilibrium): $\frac{F_1}{\sin \alpha_1} = \frac{F_2}{\sin \alpha_2} = \frac{F_3}{\sin \alpha_3}$.
- In 3D equilibrium, resolve along three axes: $\sum F_x = \sum F_y = \sum F_z = 0$ and $\sum M_x = \sum M_y = \sum M_z = 0$.

Methods

- Component along a direction: form the unit vector $\hat{u} = \vec{u} / |\vec{u}|$, then compute $\vec{a} \cdot \hat{u}$.
- Work: identify $\vec{F}$ and the displacement $\vec{d}$ (not the path), then dot them; check the sign.
- Moment in 2D: choose the pivot, draw the perpendicular distance d, then $M = Fd$ with sign (anticlockwise positive by convention).
- Moment in 3D: write $\vec{r}$ from the pivot to a point of the line of action and compute the cross product.
- Equilibrium problems: (1) draw a clear force diagram; (2) resolve in two (or three) perpendicular directions; (3) take moments about a point where unknown forces pass to eliminate them; (4) solve the system.
- Reducing a system: resultant $\vec{R} = \sum \vec{F}_i$; resultant moment $\vec{M}_O = \sum \vec{r}_i \times \vec{F}_i$. If $\vec{R} = \vec{0}$ and $\vec{M} \neq \vec{0}$, the system reduces to a couple.

Common mistakes

- Confusing scalar component (a number) with vector component (a vector).
- Using the distance along the force instead of the perpendicular distance in $M = Fd$.
- Forgetting that $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$ (order matters).
- Taking moments about different points in the same equation, or mixing sign conventions.
- For a couple, using the distance to the pivot instead of the distance between the two lines of action.
- Forgetting that work is zero when the force is perpendicular to the displacement (e.g. normal reaction on a horizontal road).
- Omitting units: work in joules (J), moment in N m — these are not the same physical quantity.

GCE tips

- Always state your sign convention for moments before writing equations.
- Choose the pivot at the point of application of an unknown reaction: it disappears from the moment equation.
- Show the force diagram: method marks are awarded for it at the GCE.
- For three-force equilibrium, Lami's theorem is often the fastest route — but check the forces are concurrent first.
- In 3D problems, work column by column with components (x, y, z) to avoid sign slips in cross products.
- Give final answers with correct units and sensible accuracy (2 or 3 significant figures).

Signed • OnBuch+
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